Tuesday recap

Kahoot-style quiz

12 quick questions on Tuesday's Mendel vocab and crosses
Not graded
Fast and correct wins.

Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-tg1-quiz](greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-tg1-quiz)

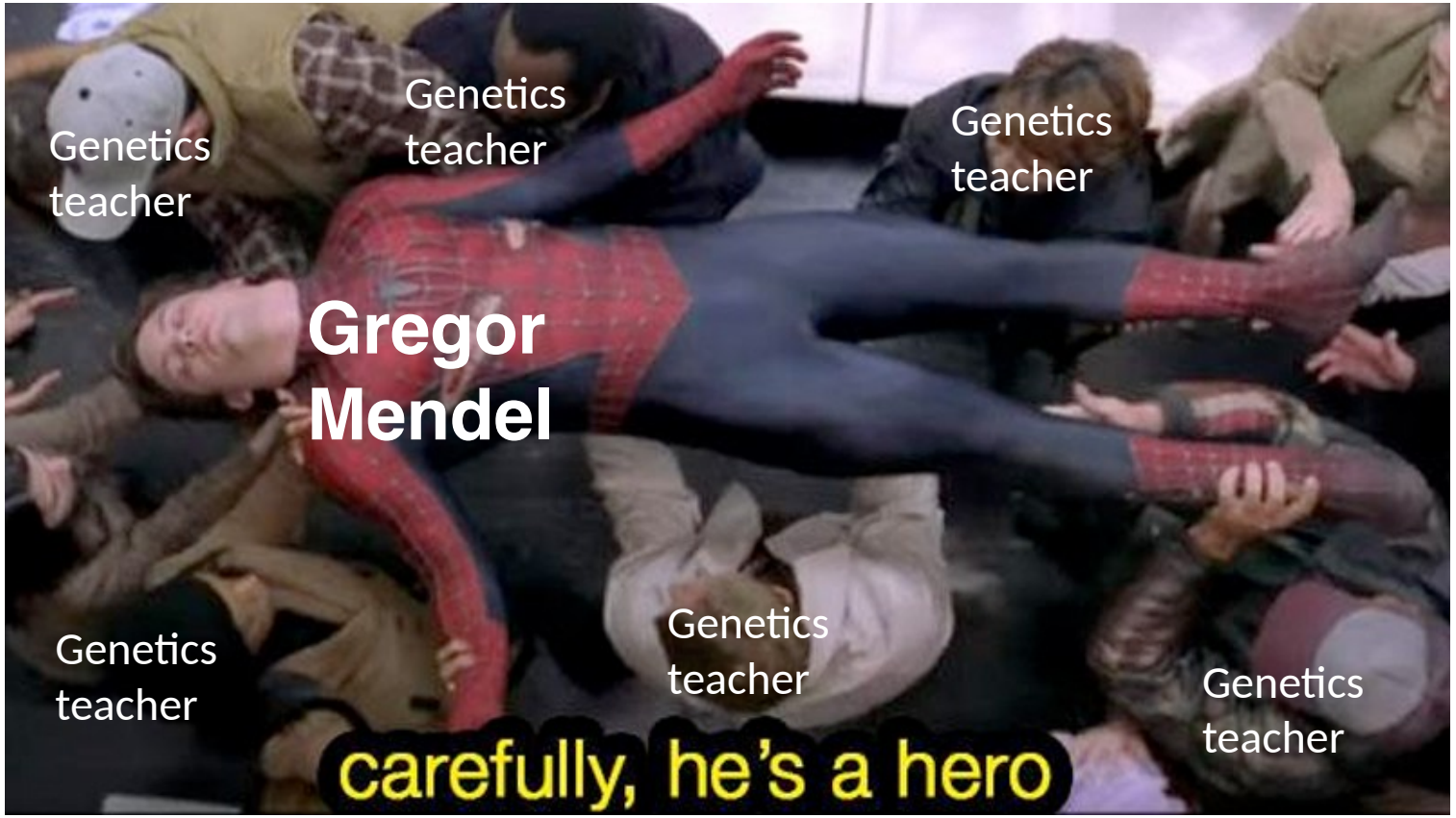
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Transmission Genetics 2: Breaking Mendel's Rules

~~Dominance~~

~~Independent assortment of genes~~

~~Random segregation of alleles~~





1. **Epistasis**: A form of gene interaction where one gene masks or modifies the effect of another gene at a different locus.
2. **Recessive epistasis**: When the recessive allele of one gene masks the effects of either allele of another gene.
3. **Dominant Epistasis**: When the dominant allele of one gene masks the effects of either allele of another gene.
4. **Synthetic lethality**: A situation where the combination of mutations in two or more genes leads to death, whereas a mutation in any single gene does not.
5. **Heterosis**: Also known as hybrid vigor, refers to the phenomenon where hybrid offspring exhibit superior phenotypic characteristics compared to their parents.
6. **Pleiotropy**: The phenomenon where a single gene affects multiple, seemingly unrelated phenotypic traits.
7. **Linkage (disequilibrium)**: The tendency of genes located close together on the same chromosome to be inherited together.
8. **Recombination**: The process by which genetic material is physically mixed during sexual reproduction, leading to the production of offspring with combinations of traits that differ from either parent.
9. **Chi-squared test**: A statistical test used to determine if observed data fits expected frequencies, often used in genetics to test for independent assortment or linkage.
10. **Marker**: A known DNA sequence or gene on a chromosome that can be used to identify a particular location or linked genes.
11. **Incomplete dominance**: A genetic situation in which one allele does not fully dominate another allele, resulting in an intermediate phenotype.
12. **Additive genetic variation**: The portion of phenotypic variation in a population that is due to the additive effects of alleles (i.e., the sum of individual effects).
13. **Co-dominance**: A genetic scenario where both alleles for a gene are expressed in a heterozygous individual, leading to a phenotype that shows both traits simultaneously.
14. **Overdominance**: A situation where the heterozygote has a phenotype that is more extreme or better adapted than either homozygote.
15. **Heterozygous advantage**: When heterozygous individuals have a higher fitness than either type of homozygous individual.
16. **Lethal alleles**: Alleles that cause the death of the organism that carries them, often before the organism can reproduce.
17. **Meiotic drive**: A phenomenon where certain alleles are transmitted to offspring more frequently than would be expected by Mendelian inheritance.
18. **Gene drive**: A genetic engineering technology used to spread a particular genetic modification throughout a population rapidly, by ensuring that a particular allele is preferentially transmitted to offspring.

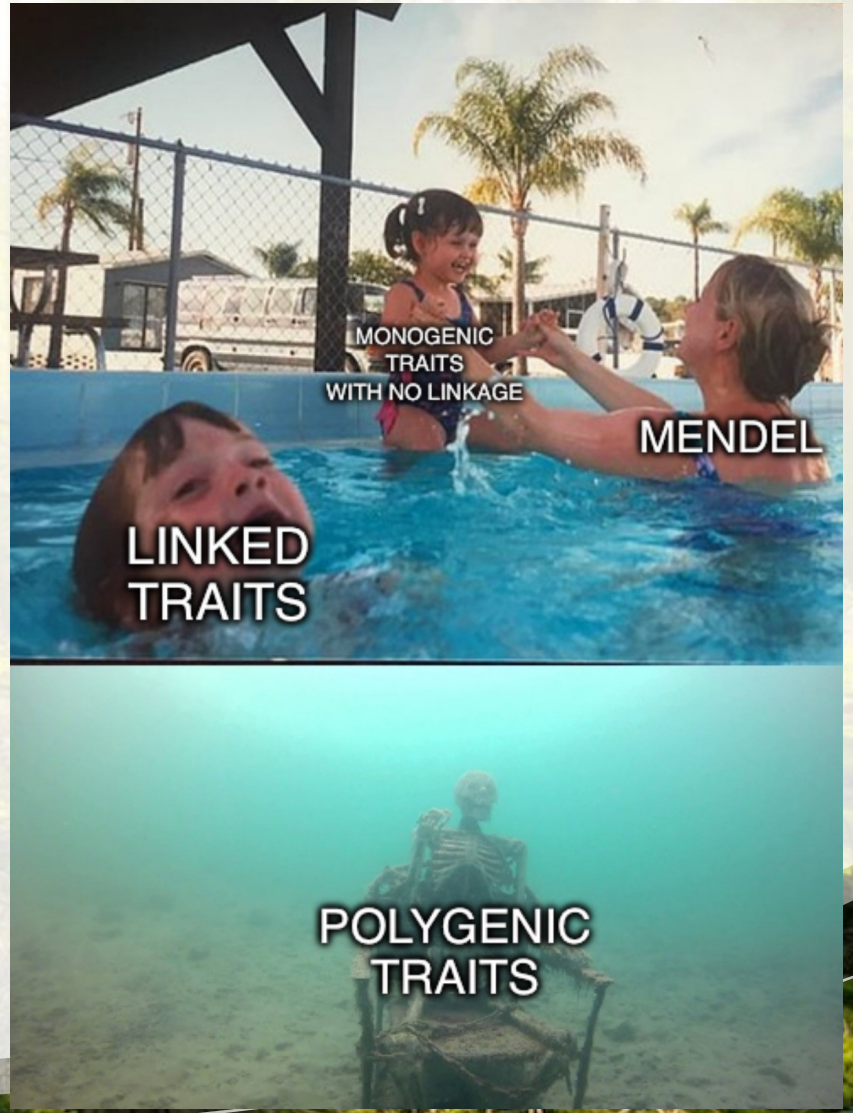
Learning goals:

The causes and consequences of non-dominant inheritance.

How gene interactions leads to epistasis.

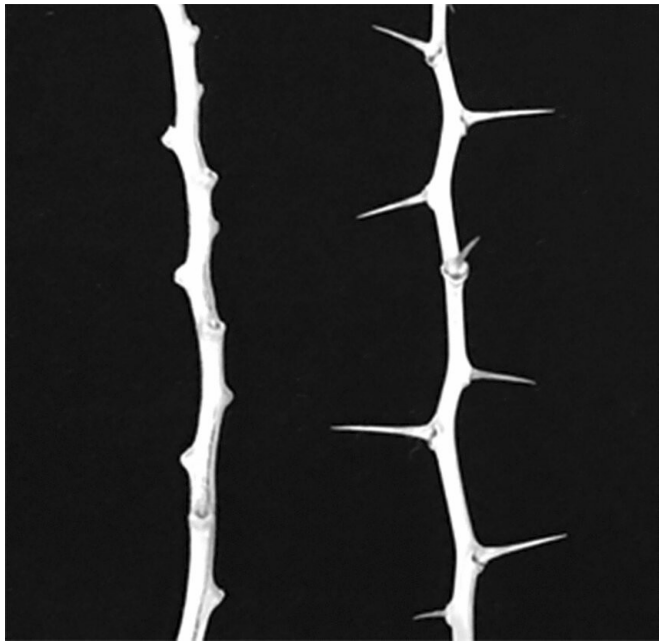
How linkage on the same chromosome violates independent assortment.

…

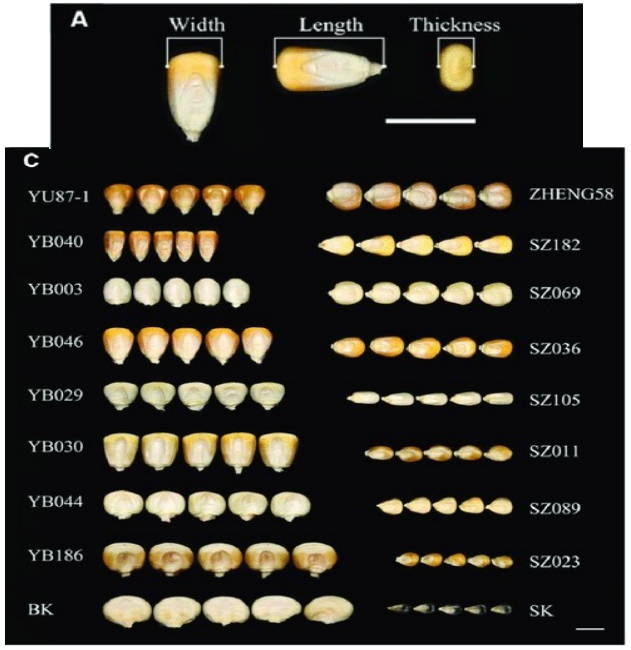


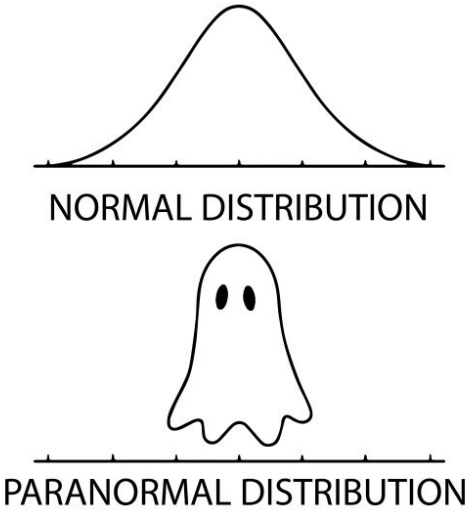
Qualitative vs quantitative traits

Qualitative (e.g. thorns absent vs present)



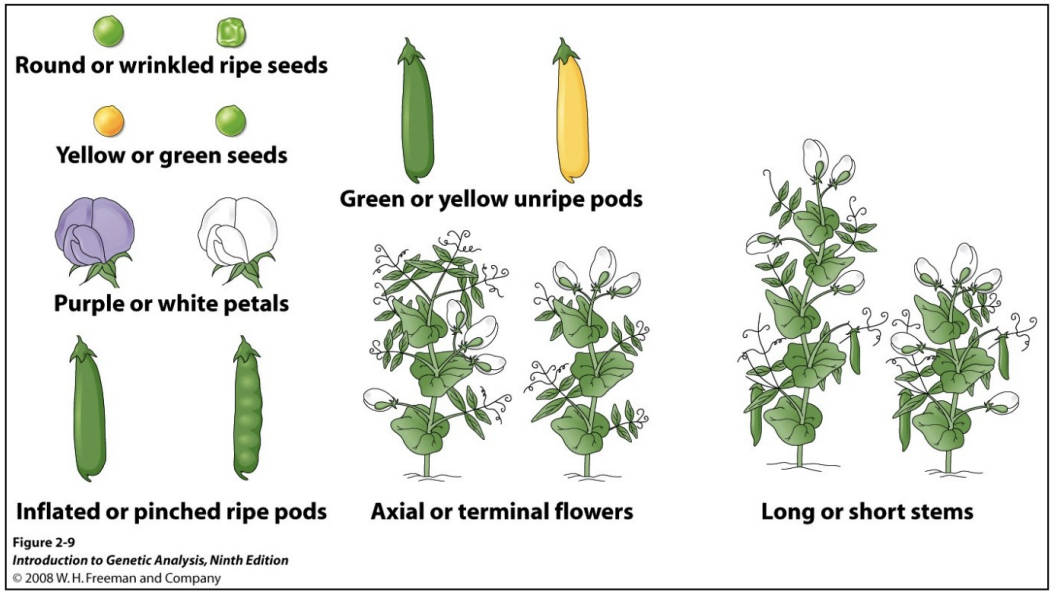
Quantitative (e.g. Kernel length)



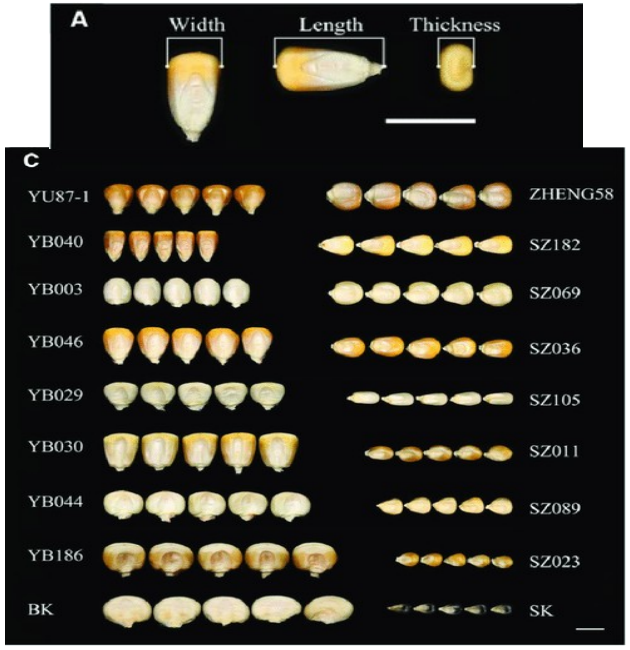


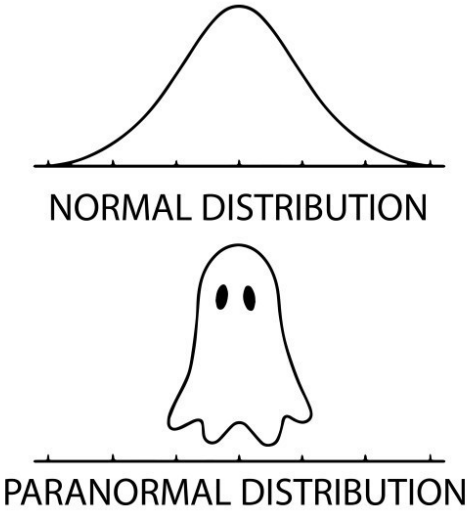
Qualitative vs quantitative traits

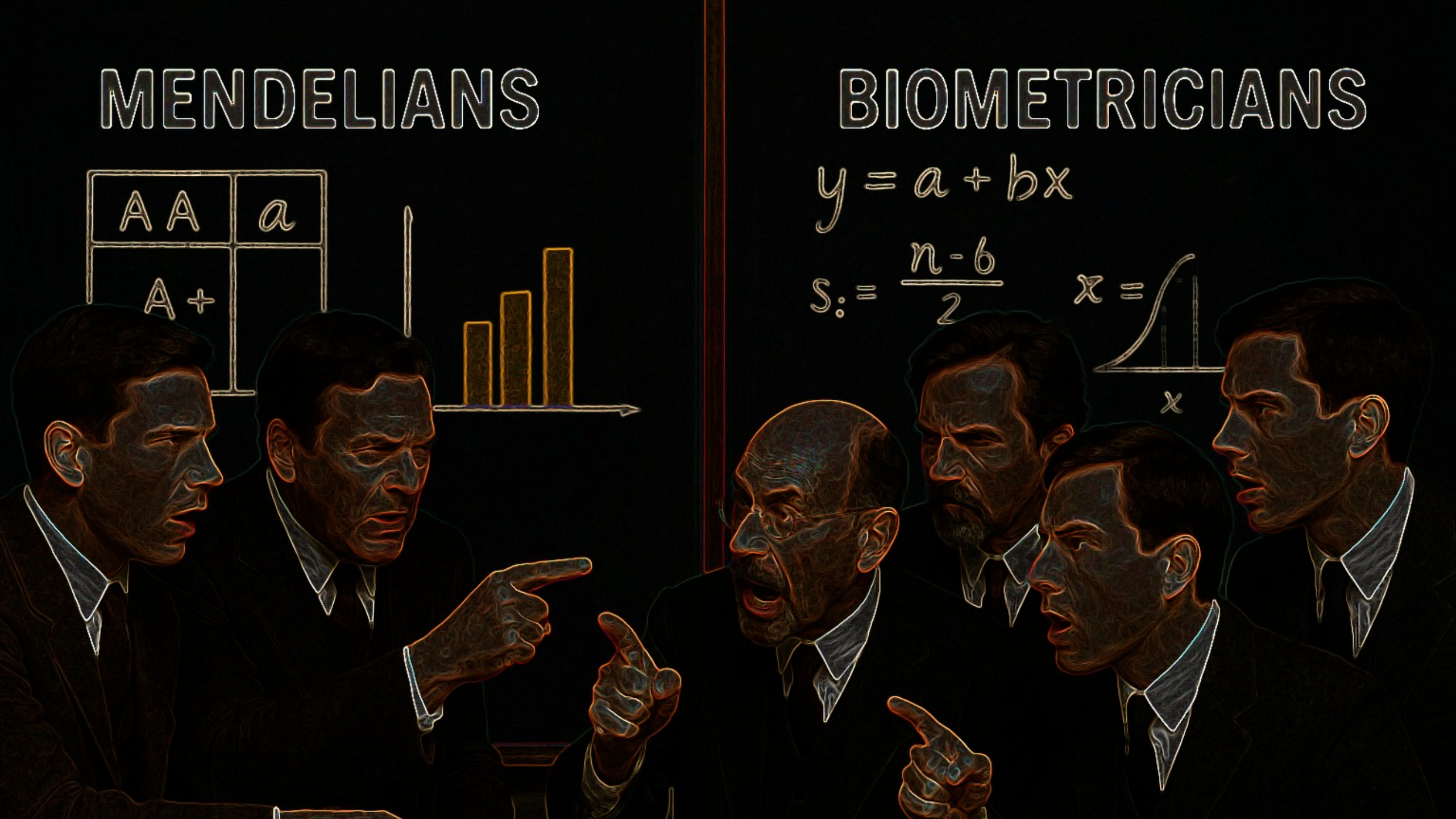
Qualitative (e.g. thorns absent vs present)



Quantitative (e.g. Kernel length)






MENDELIANS
AA
a
A+
BIOMETRICIANS
y = a + bx
s: = n - 6 / 2
x = ∫ 1 / x

MENDELIANS Conflict? BIOMETRICIANS

Qualitative “Mendelian” trait

Parent 1

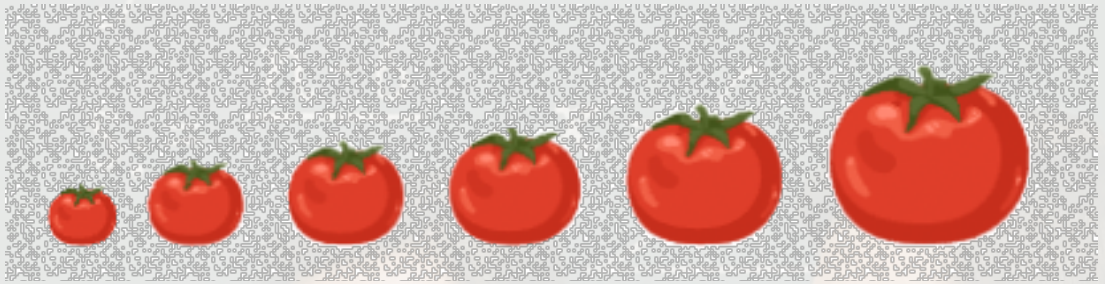


Parent 2

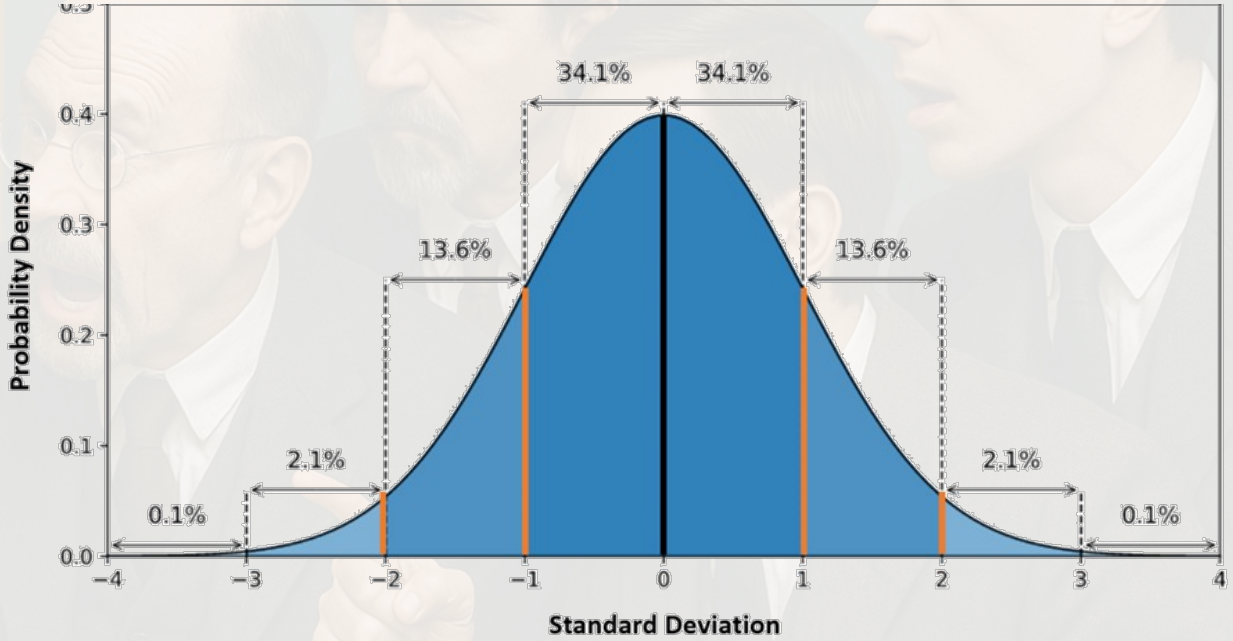


X





Quantitative trait



Live poll

Word cloud

Think of as many as you can: traits you could measure in a plant

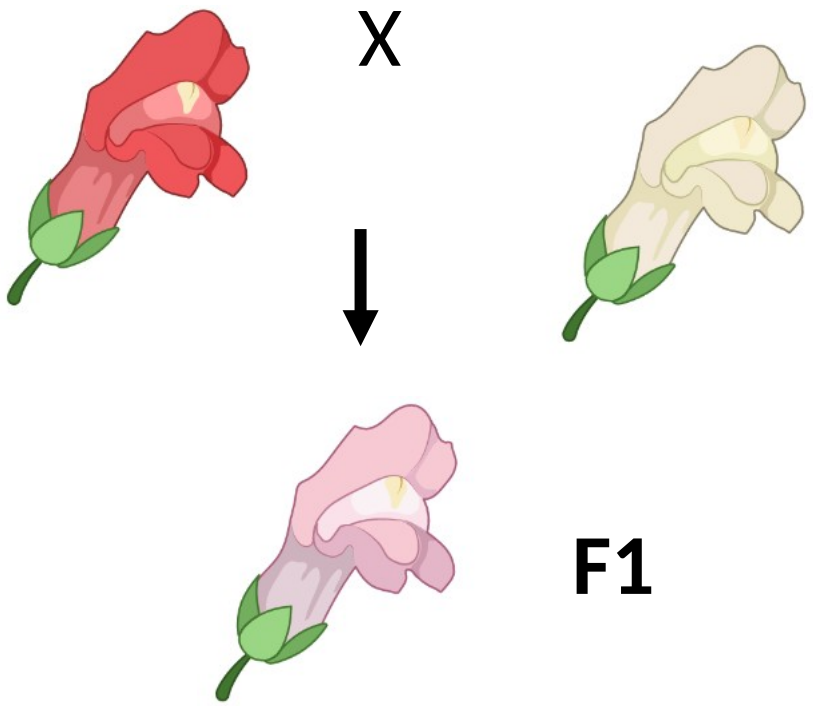
Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-traits](greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-traits)

[host screen]

Incomplete dominance

Snapdragons



Incomplete dominance



Blending inheritance?!

Snapdragons



X





F1

Incomplete dominance



Blending inheritance?!

Why is this different from blending inheritance?

Snapdragons



X







F1



25%



50%



25%

F2

Live poll

Text wall

Why is incomplete dominance different from blending inheritance?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-notblending

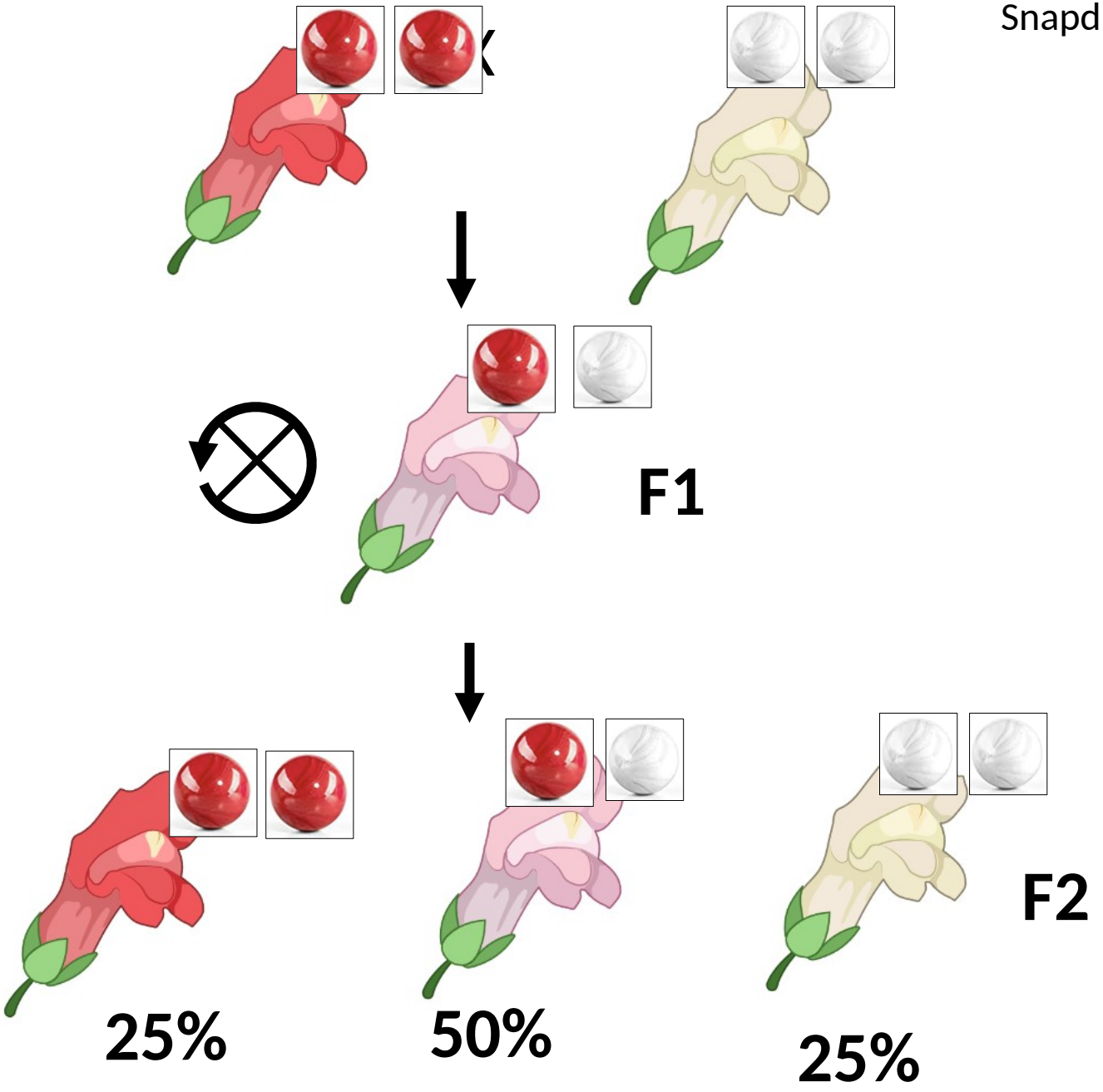
host screen

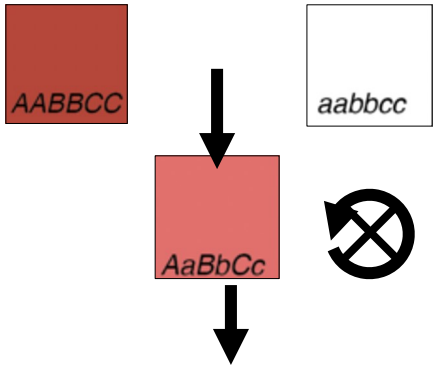
Incomplete dominance



Why is this different from blending inheritance?

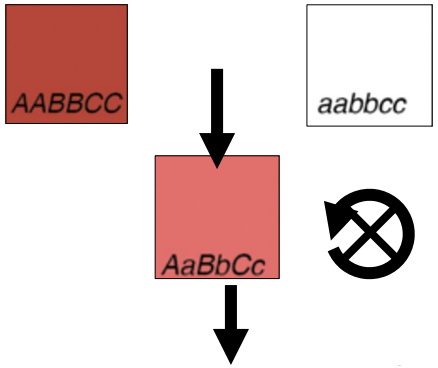
Snapdragons





Each uppercase allele (A, B, C) contributes 1 unit of pigment

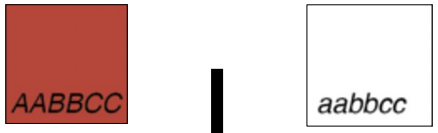
Incomplete dominance in many genes leads to "additive genetic variation"

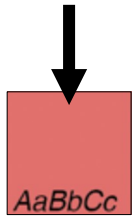


Each uppercase allele (A,B,C) contributes 1 unit of pigment

	ABC	AbC	aBC	ABc	Abc	abC	aBc	abc
ABC	AABBCC	AABbCC	AaBBCC	AABBCc	AABbCc	AaBbCC	AaBBCc	AaBbCc
AbC	AABbCC	AAbbCC	AaBbCC	AABbCc	AAbbCc	AabbCC	AaBbCc	AabbCc
aBC	AaBBCC	AaBbCC	aaBBCC	AaBBCc	AaBbCc	aaBbCC	aaBBCc	aaBbCc
ABc	AABBCc	AABbCc	AaBBCc	AABBcc	AABbcc	AaBbCc	AaBBcc	AaBbcc
Abc	AABbCc	AAbbCc	AaBbCc	AABbcc	AAbbcc	AabbCc	AaBbcc	Aabbcc
abC	AaBbCC	AabbCC	aaBbCC	AaBbCc	AabbCc	aabbCC	aaBbCc	aabbCc
aBc	AaBBCc	AaBbCc	aaBBCc	AaBBcc	AaBbcc	aaBbCc	aaBBcc	aaBbcc
abc	AaBbCc	AabbCc	aaBbCc	AaBbcc	Aabbcc	aabbCc	aaBbcc	aabbcc

Incomplete dominance in many genes leads to “additive genetic variation”



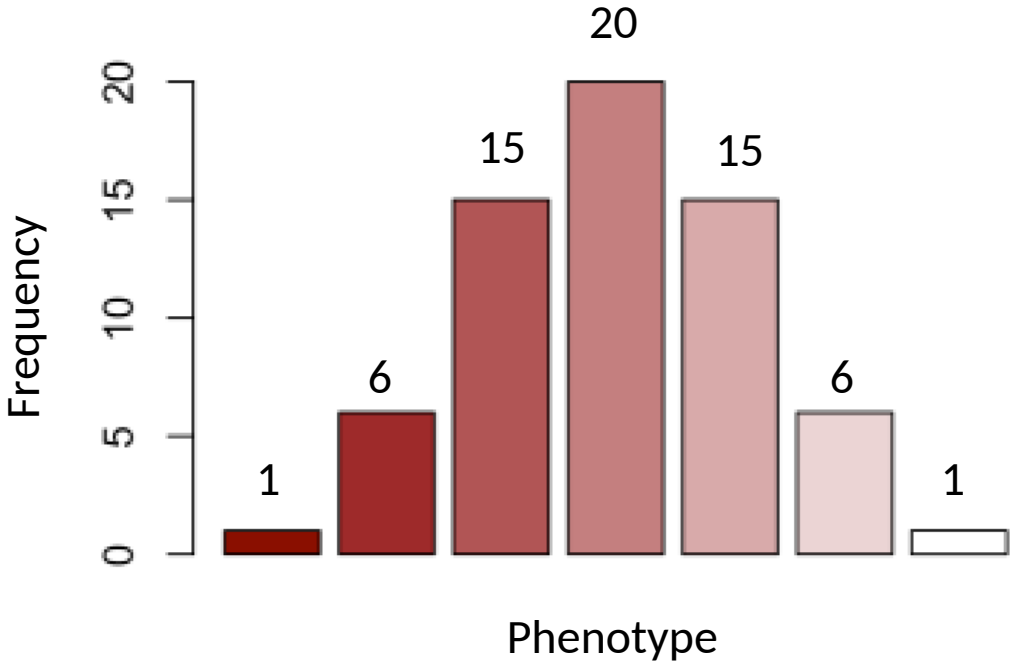




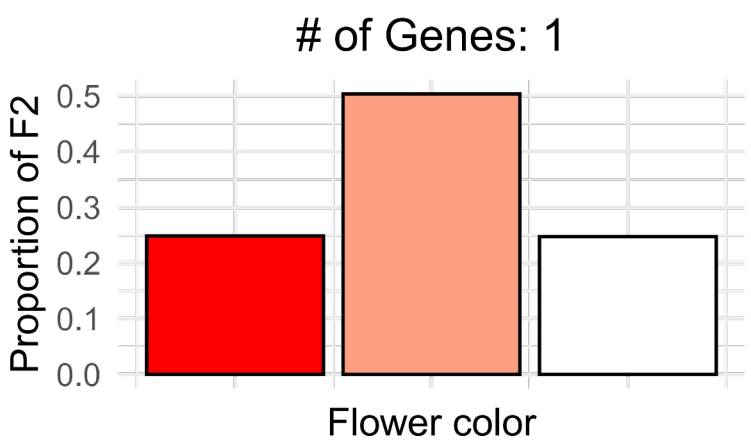
Each uppercase allele (A,B,C) contributes 1 unit of pigment

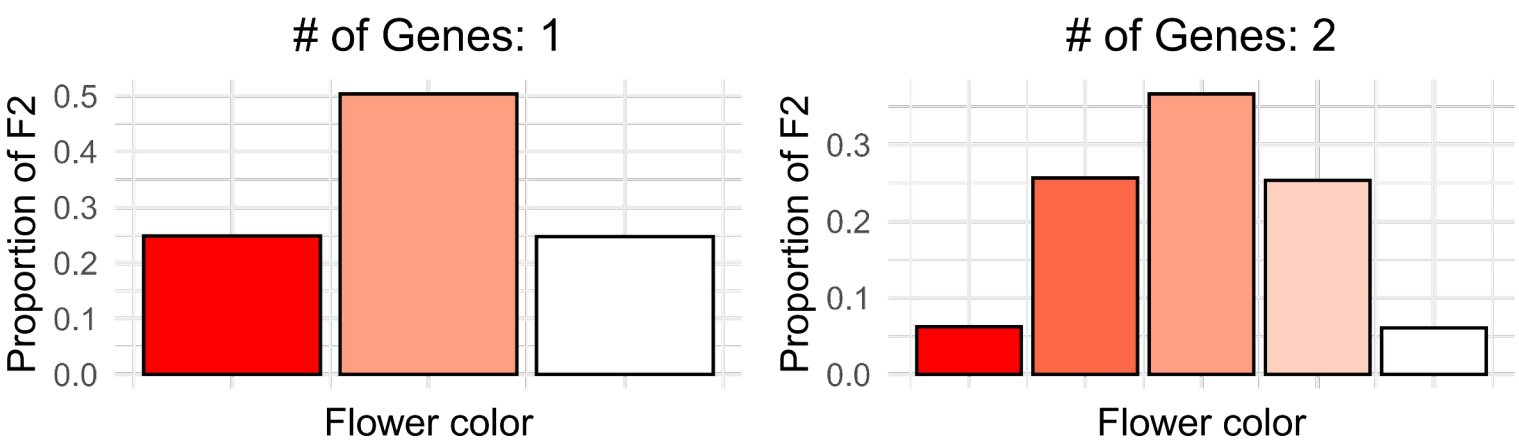
Incomplete dominance in many genes leads to "additive genetic variation"

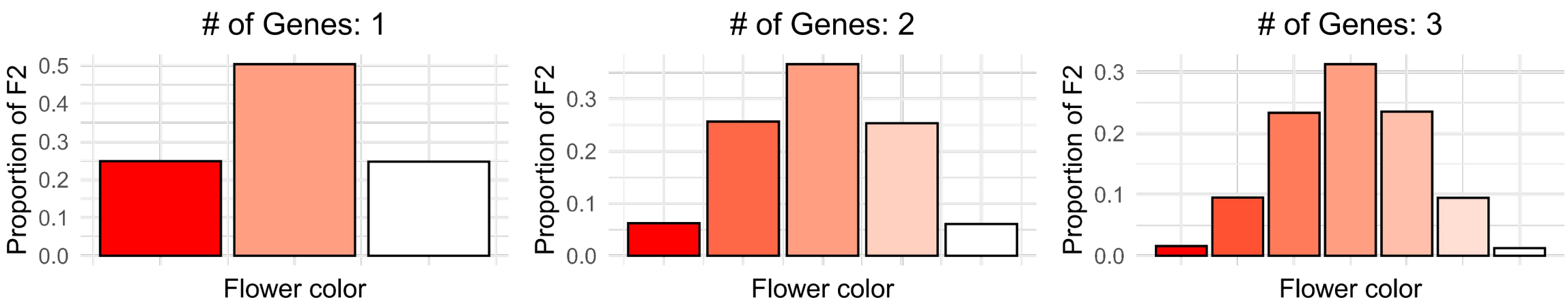
	ABC	AbC	aBC	ABc	Abc	abC	aBc	abc
ABC	AABBCC	AABbCC	AaBBCC	AABBCc	AABbCc	AaBbCC	AaBBCc	AaBbCc
AbC	AABbCC	AAbbCC	AaBbCC	AABbCc	AAbbCc	AabbCC	AaBbCc	AabbCc
aBC	AaBBCC	AaBbCC	aaBBCC	AaBBCc	AaBbCc	aaBbCC	aaBBCc	aaBbCc
ABc	AABBCc	AABbCc	AaBBCc	AABBcc	AABbcc	AaBbCc	AaBBcc	AaBbcc
Abc	AABbCc	AAbbCc	AaBbCc	AABbcc	AAbbcc	AabbCc	AaBbcc	Aabbcc
abC	AaBbCC	AabbCC	aaBbCC	AaBbCc	AabbCc	aabbCC	aaBbCc	aabbCc
aBc	AaBBCc	AaBbCc	aaBBCc	AaBBcc	AaBbcc	aaBbCc	aaBBcc	aaBbcc
abc	AaBbCc	AabbCc	aaBbCc	AaBbcc	Aabbcc	aabbCc	aaBbcc	aabbcc

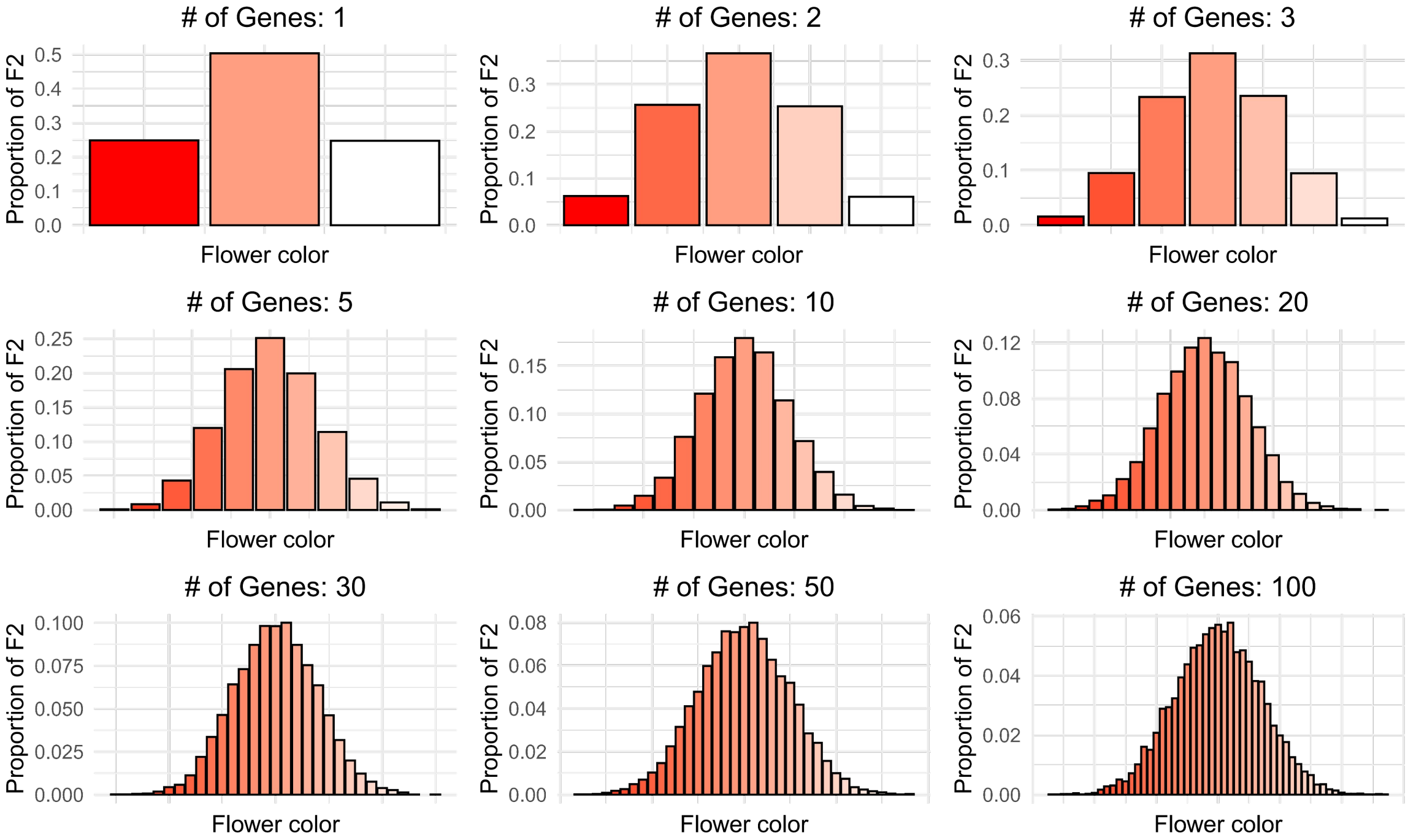


possible phenotypes = # genes * 2 + 1

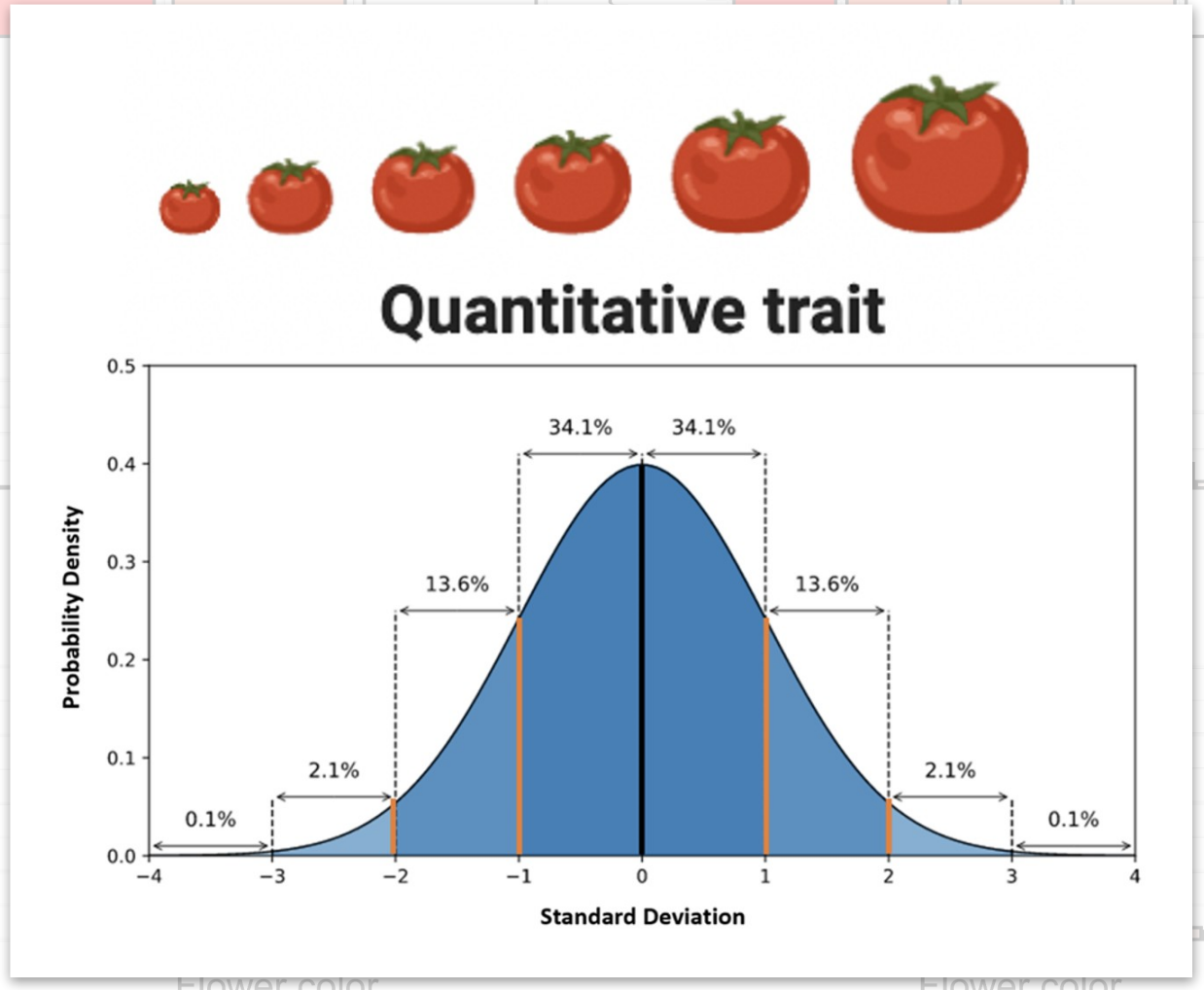

of Genes: 1
Proportion of F2
0.5
0.4
0.3
0.2
0.1
0.0
Flower color

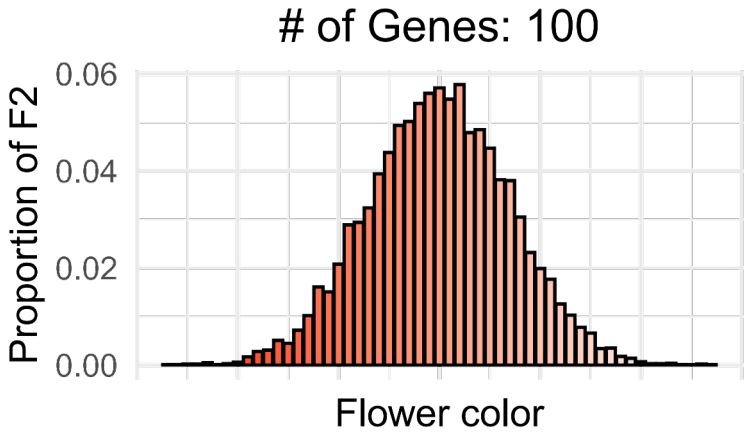

of Genes: 1
Proportion of F2
0.5
0.4
0.3
0.2
0.1
0.0
Flower color
of Genes: 2
Proportion of F2
0.3
0.2
0.1
0.0
Flower color

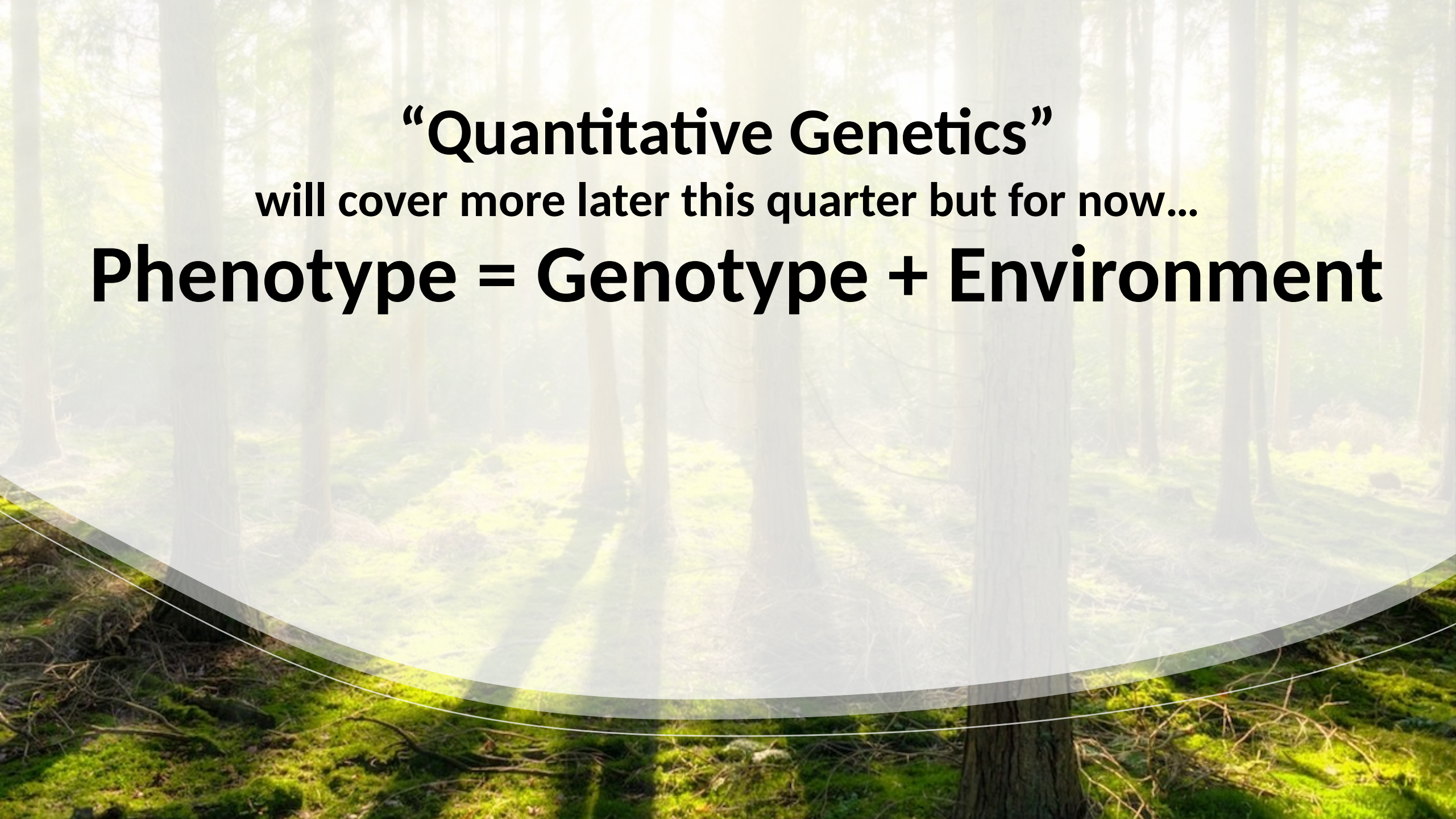

of Genes: 1
Proportion of F2
0.5
0.4
0.3
0.2
0.1
0.0
Flower color
of Genes: 2
Proportion of F2
0.3
0.2
0.1
0.0
Flower color
of Genes: 3
Proportion of F2
0.3
0.2
0.1
0.0
Flower color


of Genes: 1
Proportion of F2
0.0
0.1
0.2
0.3
0.4
0.5
Flower color
of Genes: 2
Proportion of F2
0.0
0.1
0.2
0.3
Flower color
of Genes: 3
Proportion of F2
0.0
0.1
0.2
0.3
Flower color
of Genes: 5
Proportion of F2
0.00
0.05
0.10
0.15
0.20
0.25
Flower color
of Genes: 10
Proportion of F2
0.00
0.05
0.10
0.15
Flower color
of Genes: 20
Proportion of F2
0.00
0.04
0.08
0.12
Flower color
of Genes: 30
Proportion of F2
0.000
0.025
0.050
0.075
0.100
Flower color
of Genes: 50
Proportion of F2
0.00
0.02
0.04
0.06
0.08
Flower color
of Genes: 100
Proportion of F2
0.00
0.02
0.04
0.06
Flower color

Quantitative traits are consistent with Mendelian genetics if polygenic: many genes contribute to the trait







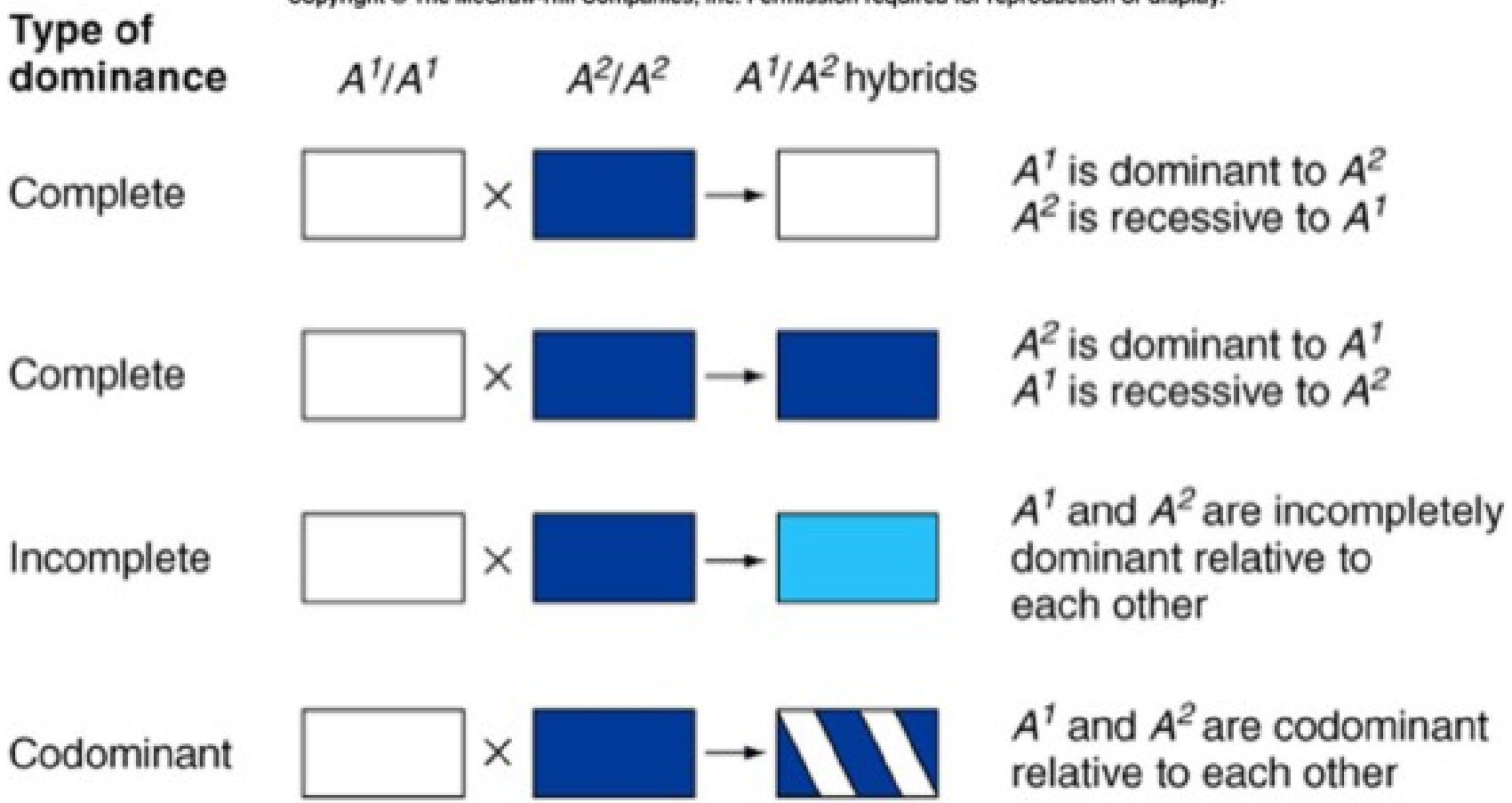
“Quantitative Genetics”

will cover more later this quarter but for now…

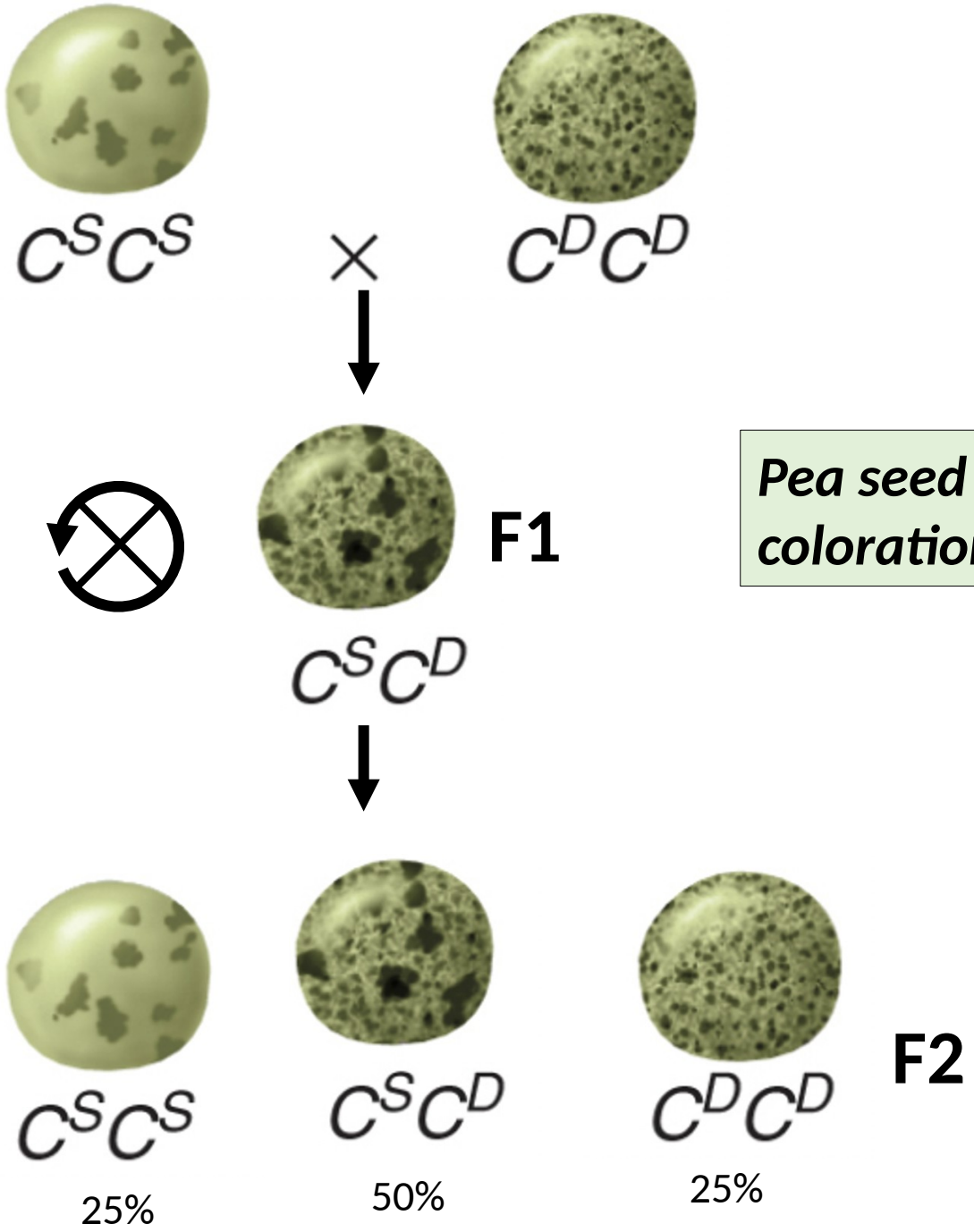
Phenotype = Genotype + Environment

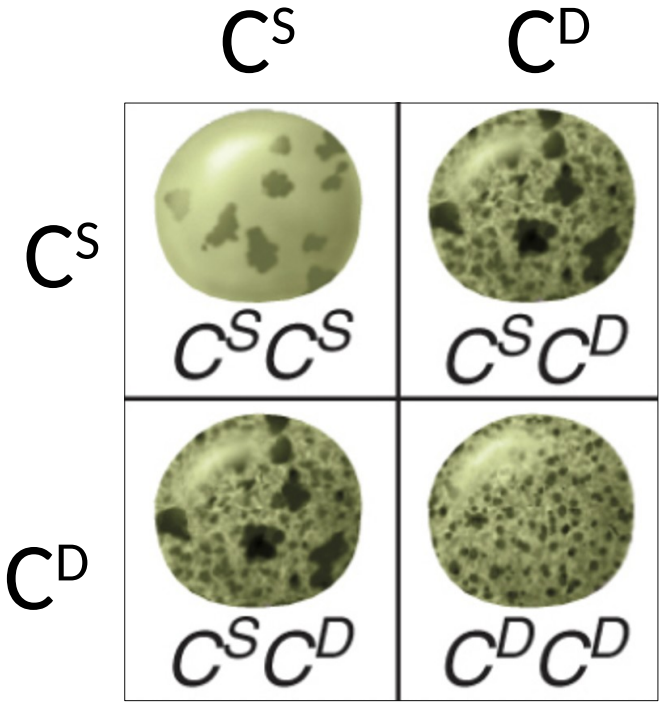
Summary: types of dominance

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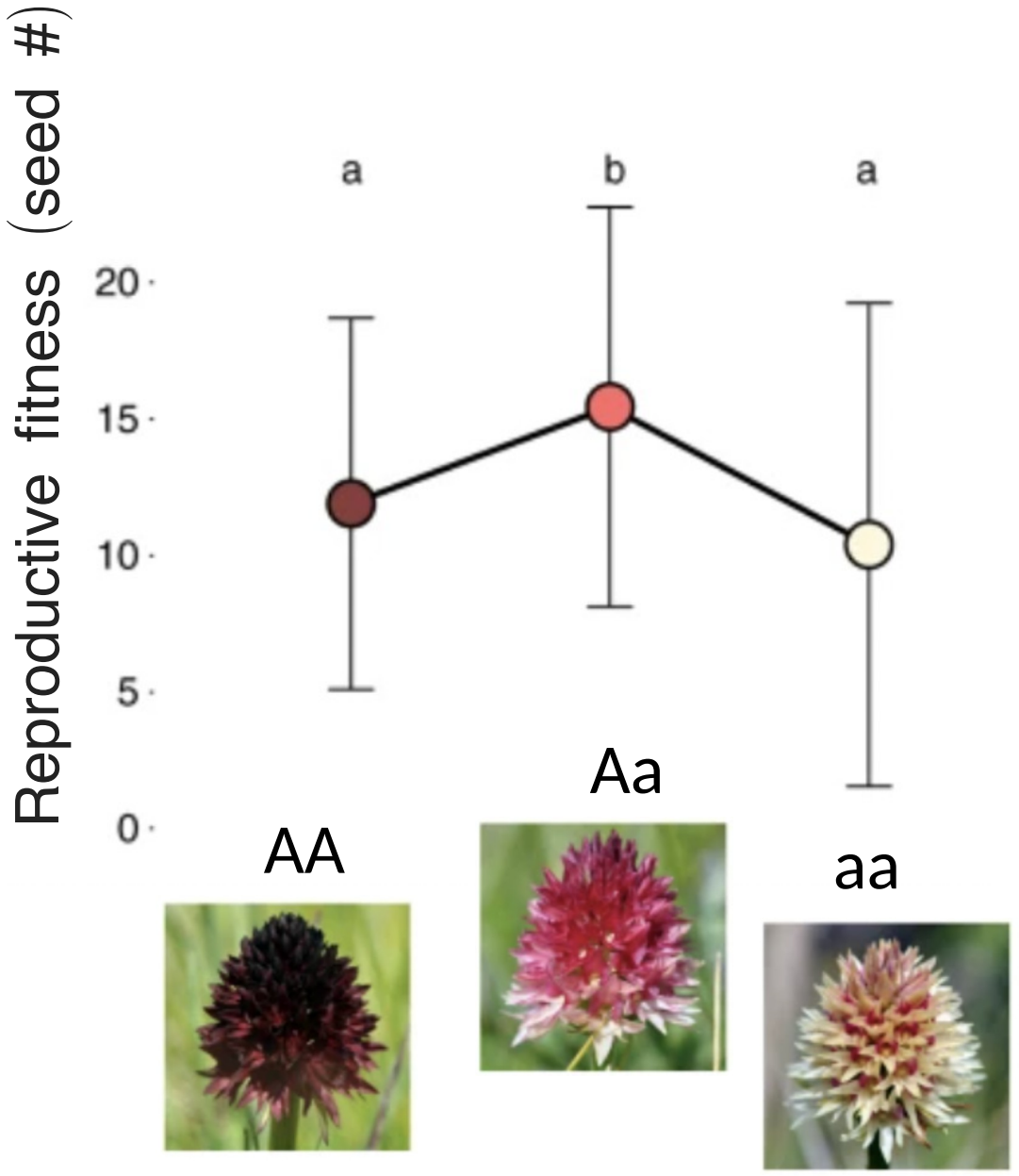
Co-dominance: both alleles show phenotype





Overdominance:
best of both worlds aka
"**heterozygote advantage**"

Why might the heterozygote have better fitness?



Alpine orchid *Gymnadenia rhellicani*

Live poll

Text wall

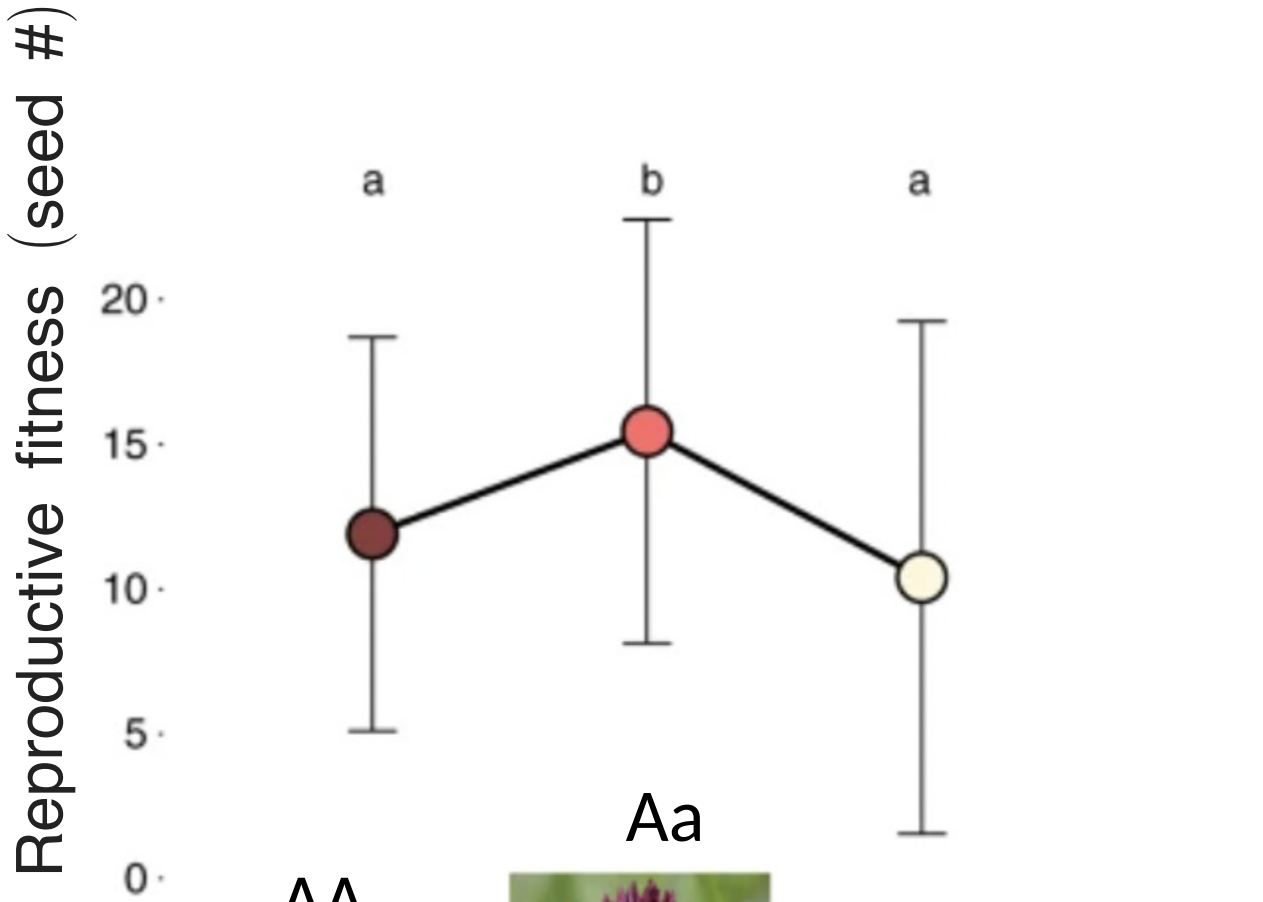
Why might the heterozygote have better fitness?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-overdom

host screen

Overdominance:
best of both worlds aka
"**heterozygote advantage**"

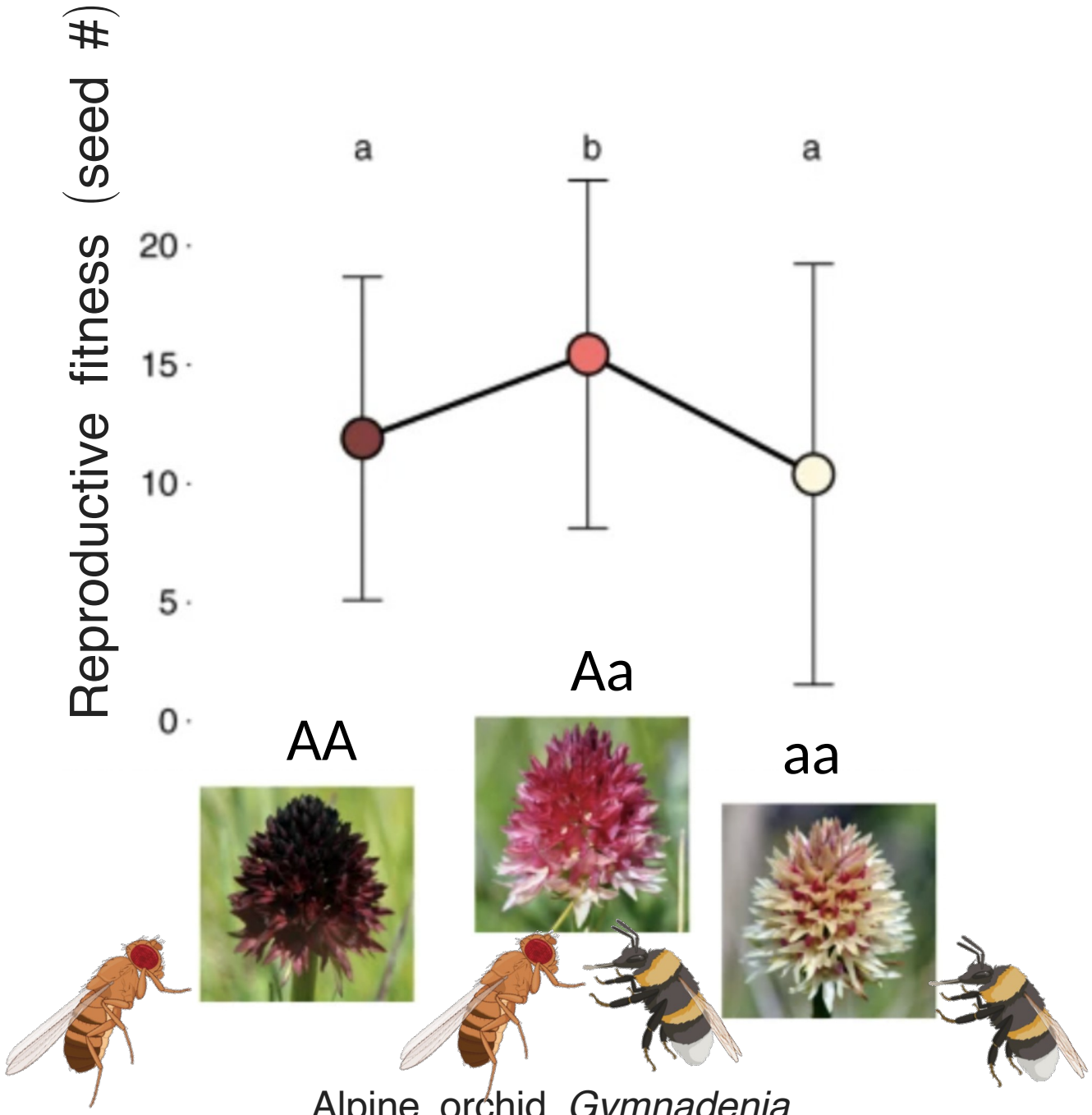


Why might the heterozygote have better fitness?

Alpine orchid *Gymnadenia rhellicani*

Overdominance:
best of both worlds aka
"**heterozygote advantage**"

Why might the heterozygote have better fitness?



Alpine orchid *Gymnadenia rhellicani*

Disease resistance A/A
b/b



X



a/a
B/B **Ear size**





F1

Hybrid – seed sold to farmers

A/a
b/B





Disease resistance A/A
b/b



X

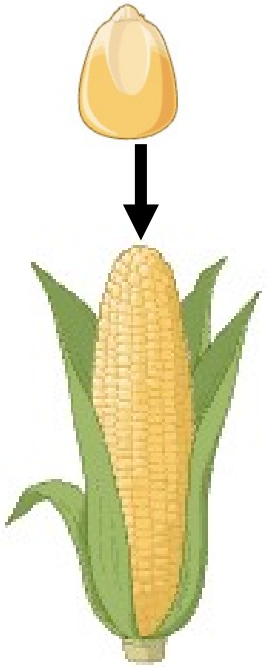


a/a
B/B **Ear size**





A/a
b/B



F1

Hybrid – seed sold to farmers



Disease resistance A/A
b/b



X



a/a
B/B **Ear size**





F1

Hybrid – seed sold to farmers

A/a
b/B







Disease resistance A/A
b/b



X



a/a
B/B **Ear size**





F1

Hybrid – seed sold to farmers

A/a
b/B



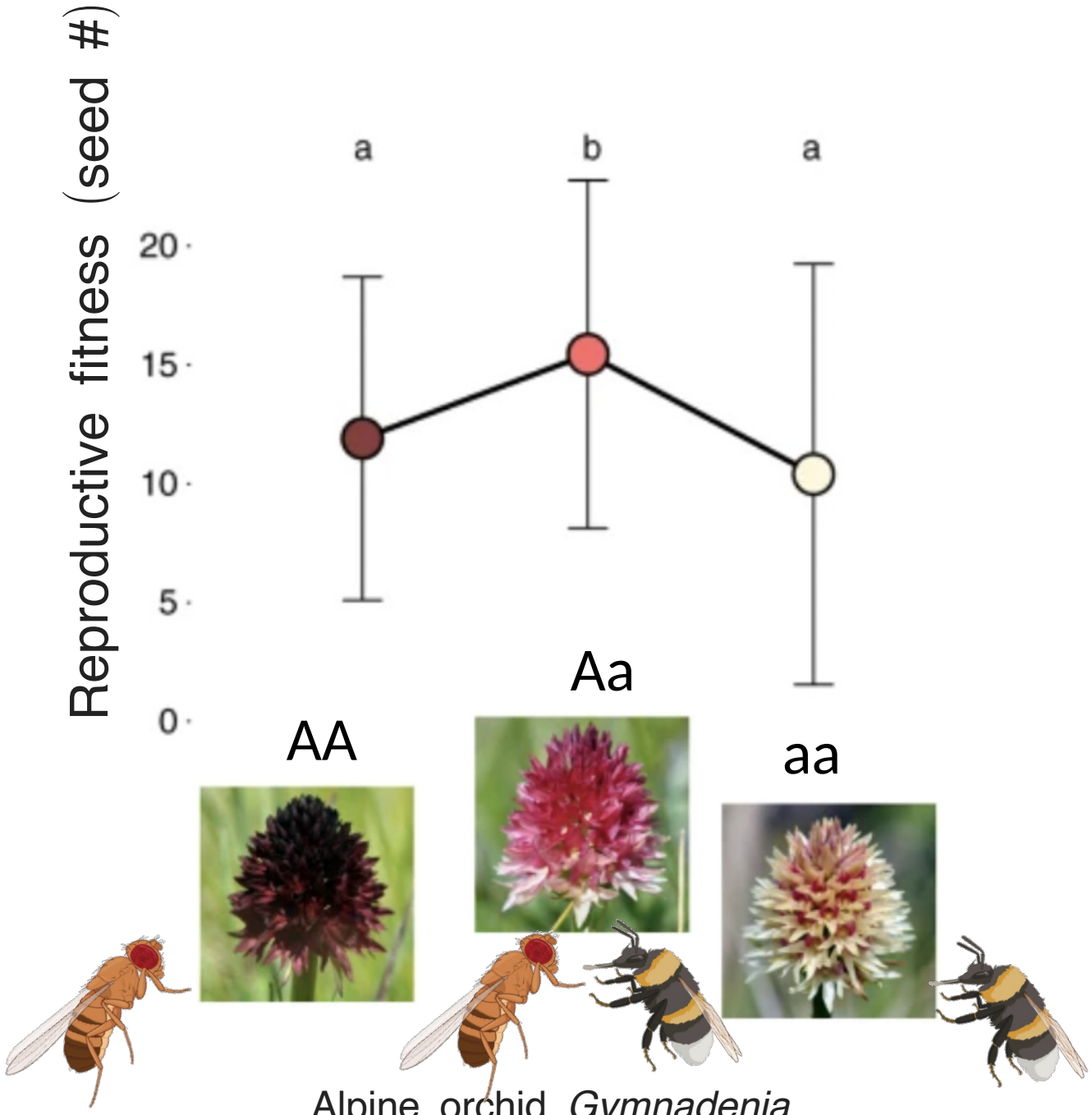




F2 will have some homozygous deleterious alleles and be less genetically uniform

Overdominance:
best of both worlds aka
"**heterozygote advantage**"

Why might the heterozygote have better fitness?



Alpine orchid *Gymnadenia rhellicani*

Heterosis (AKA hybrid vigor): overdominance?

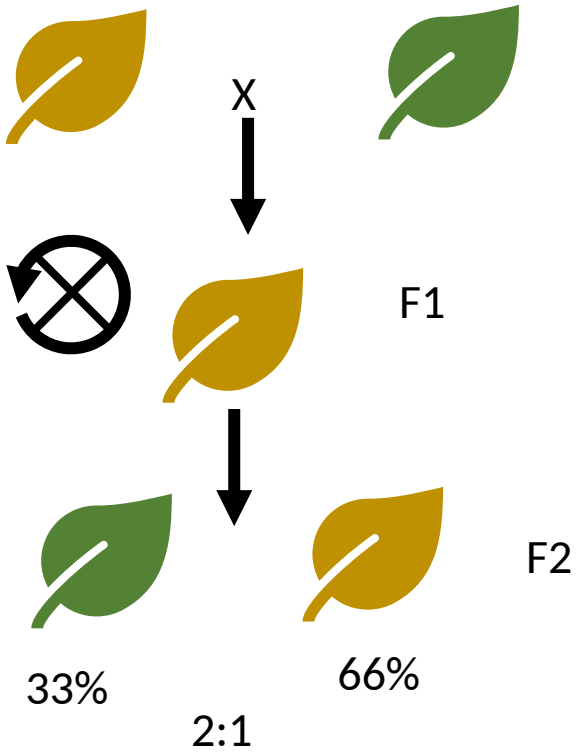


Other deviations from Mendelian expectations…



Nature Reviews | Genetics

Snapdragon



YG x YG

	Y	G
Y	YY	YG
G	YG	GG

Why might we see a 2:1 ratio in the F2?

Live poll

Text wall

Why might we see a 2:1 ratio in the F2?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-twoone

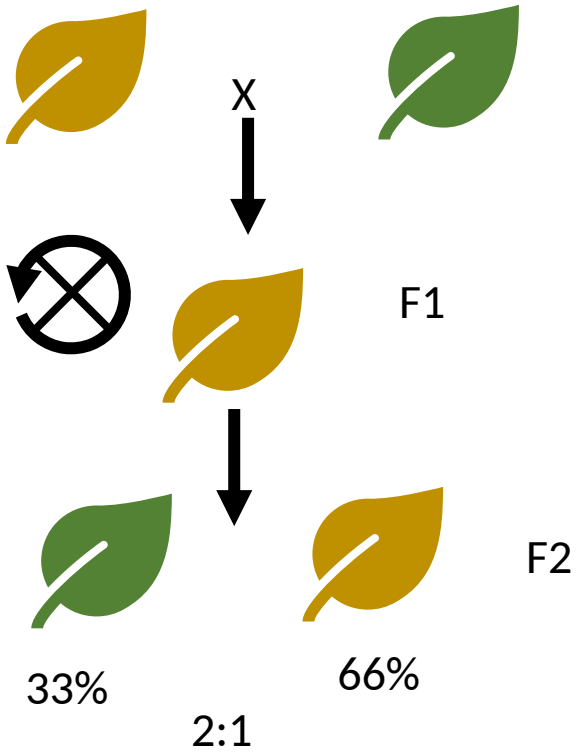
host screen

Homozygous lethal alleles



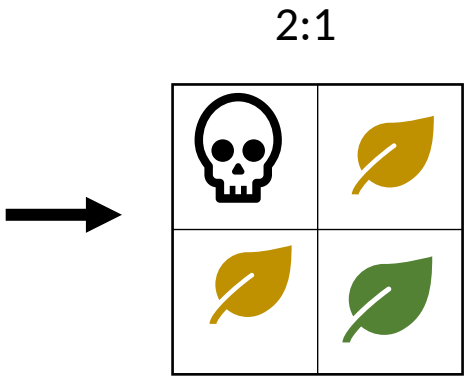
Nature Reviews | Genetics

Snapdragon



YG x YG

	Y	G
Y	YY	YG
G	YG	GG



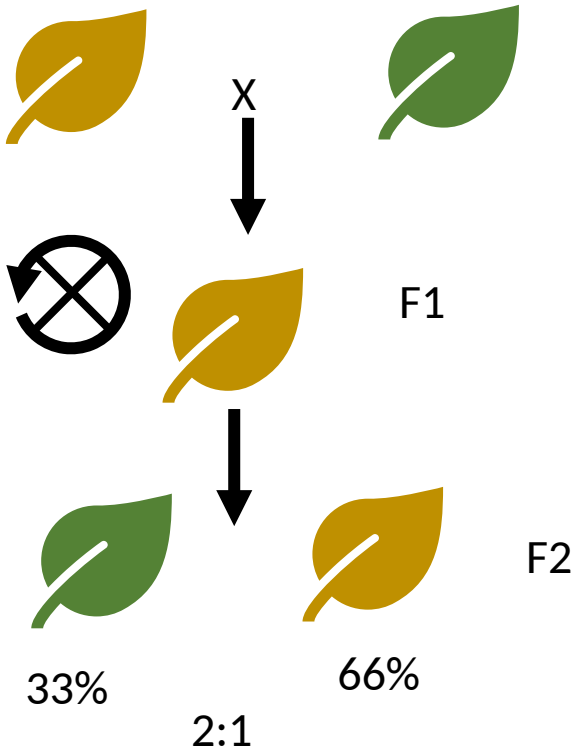
Homozygous lethal alleles



Nature Reviews | Genetics

Snapdragon

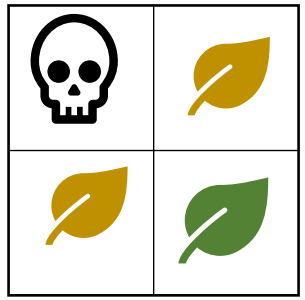
What is the genotype of this plant?



YG x YG

	Y	G
Y	YY	YG
G	YG	GG

2:1



Live poll

Multiple choice

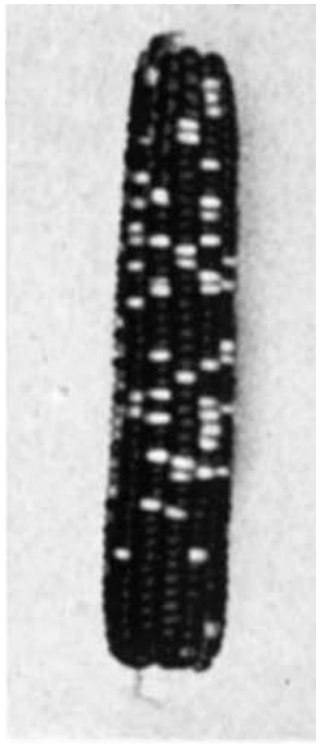
YG × YG snapdragons: what is the genotype of this yellow-green plant?

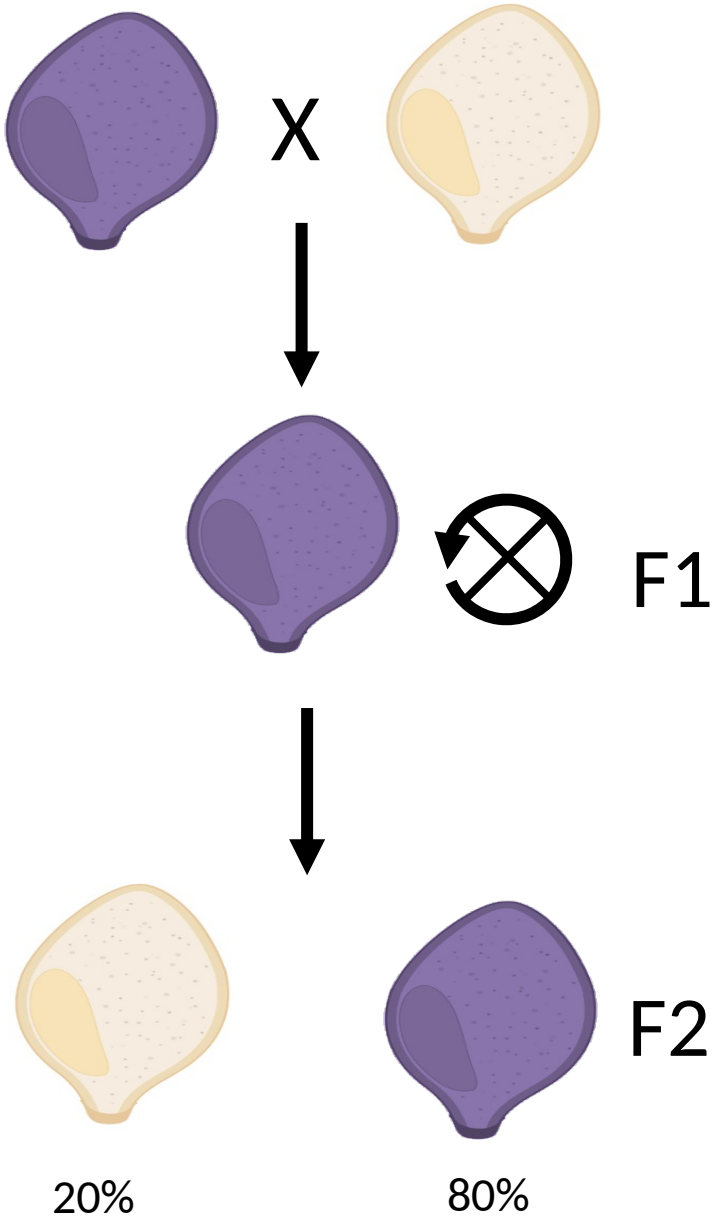
Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-lethalgeno

host screen



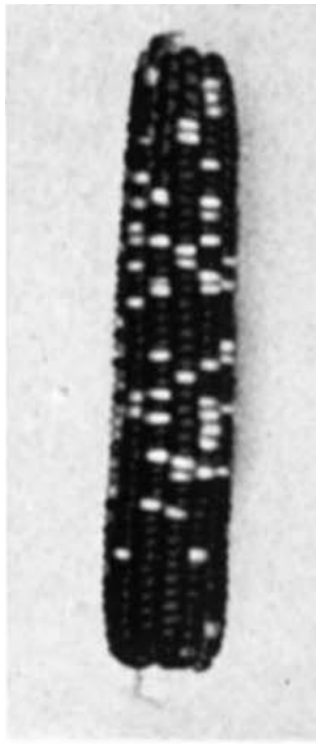


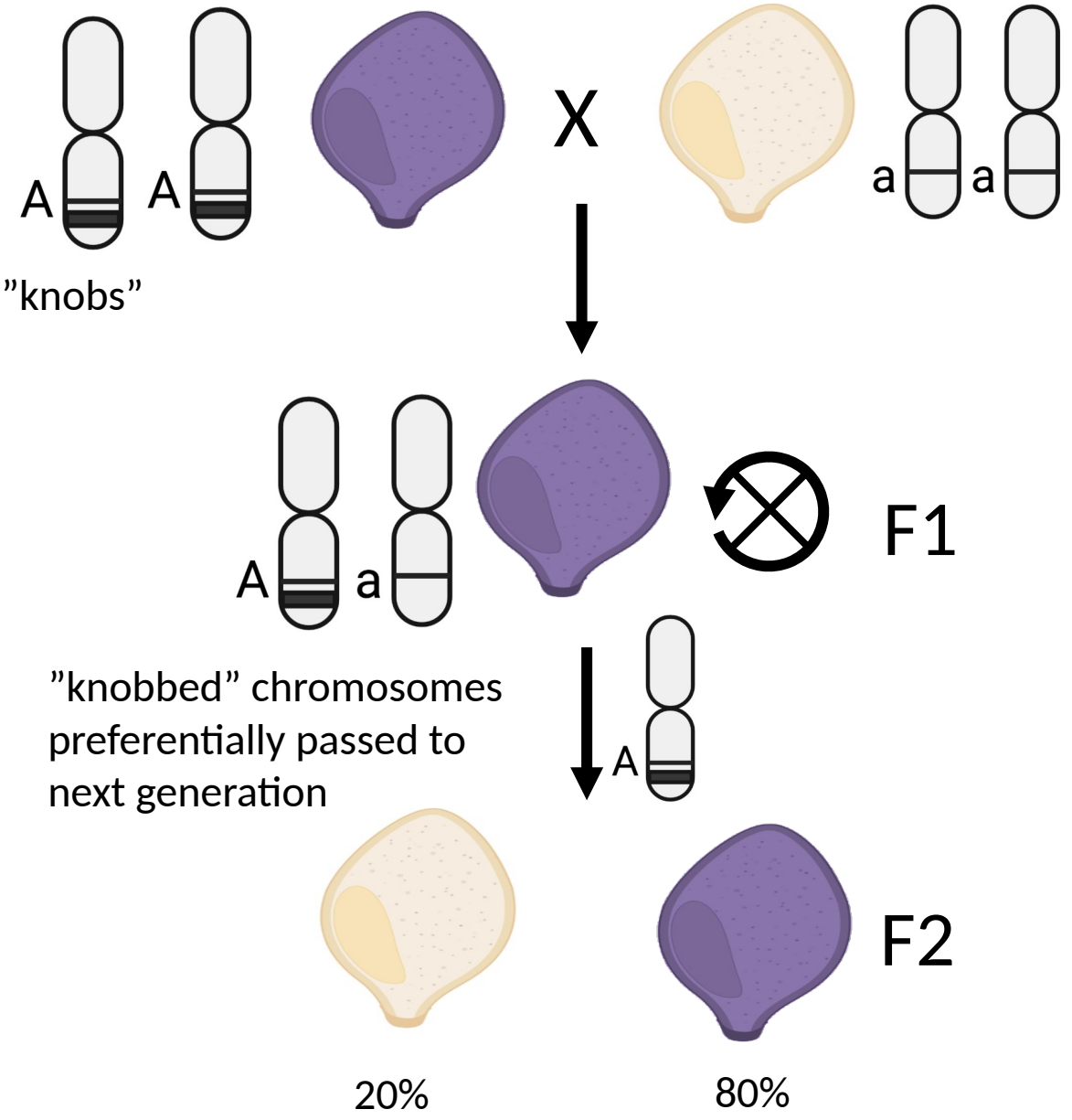


Counted thousands of corn kernels and found a clear 4:1 Ratio

Meiotic drive: one allele inherited more often than another

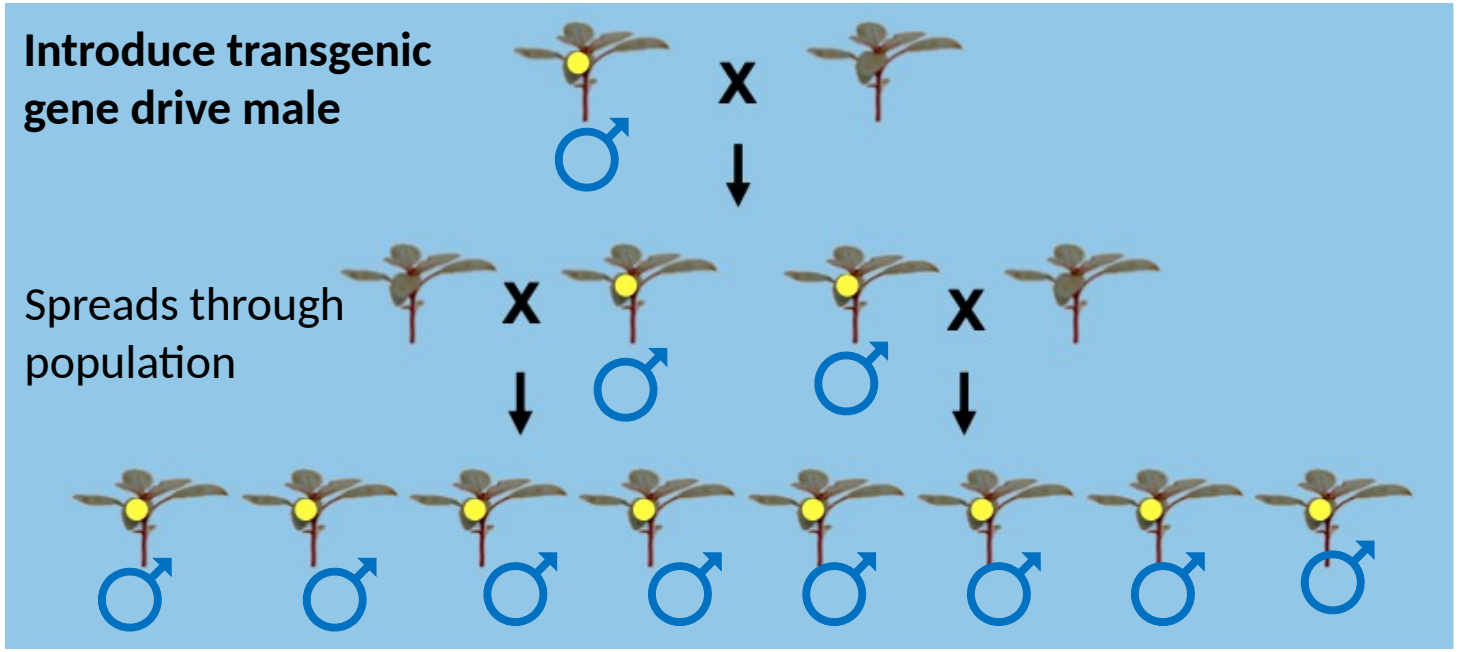






Counted thousands of corn kernels and found a clear 4:1 Ratio

Should we use meiotic drive to eliminate weeds?



Perspective

SCI

Received: 29 March 2018 Revised: 6 July 2018 Accepted article published: 12 July 2018 Published online in Wiley Online Library: 18 September 2018

(wileyonlinelibrary.com) DOI 10.1002/ps.5137

Gene drive systems: do they have a place in agricultural weed management?

Paul Neve*

Gene drives in plants: opportunities and challenges for weed control and engineered resilience

Luke G. Barrett[1,3], Mathieu Legros[1,3], Nagalingam Kumaran[2], Donna Glassop[1], S. Raghu[2] and Donald M. Gardiner[1]

[1]CSIRO Agriculture and Food, Canberra, Australian Capital Territory, Australia

Debate!

PRO vs ANTI chat wall

Meiotic drive to wipe out a weed that devastates tomato farms?
Pick a side. Post your best argument. React to the other side's posts.

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/debate-play.html?poll=pls152-tg2-drivedebate

host screen

Live poll

Multiple choice

Ballot initiative: a weed causes major harm to tomato farmers, and scientists propose using meiotic drive to eliminate it. How do you vote?

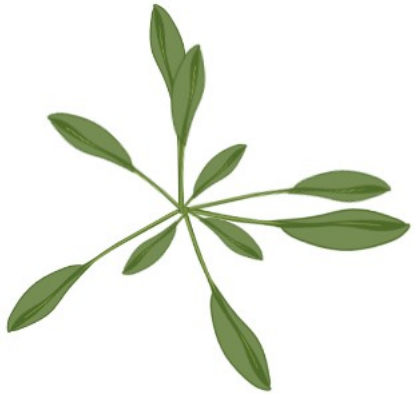
Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-drive

host screen



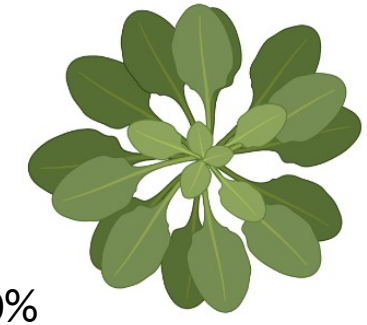
X





F1





F2

100%

Live poll

Text wall

Self the normal F1 and 100% of the F2 look normal. Where did the recessive phenotype go? Best idea wins.

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-envf2

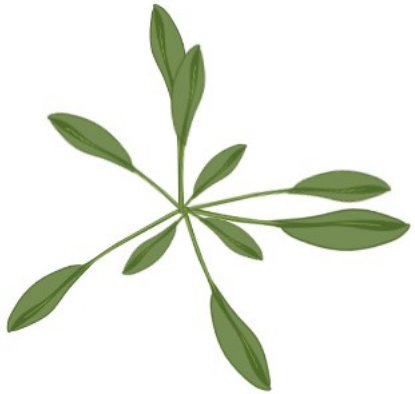
host screen

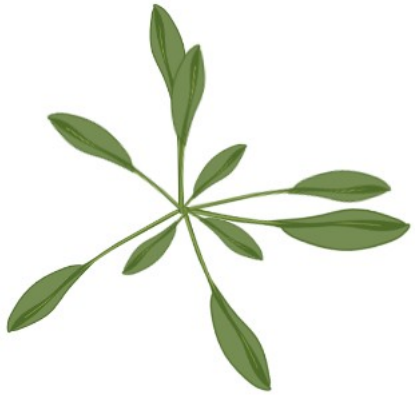
Environmental dependance





X





E.g. Heat-shock protein mutant



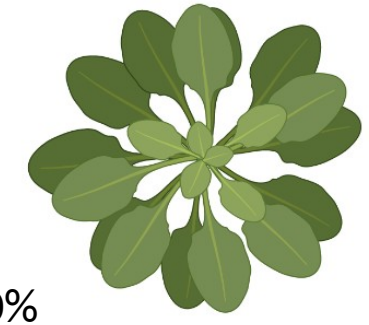




F1







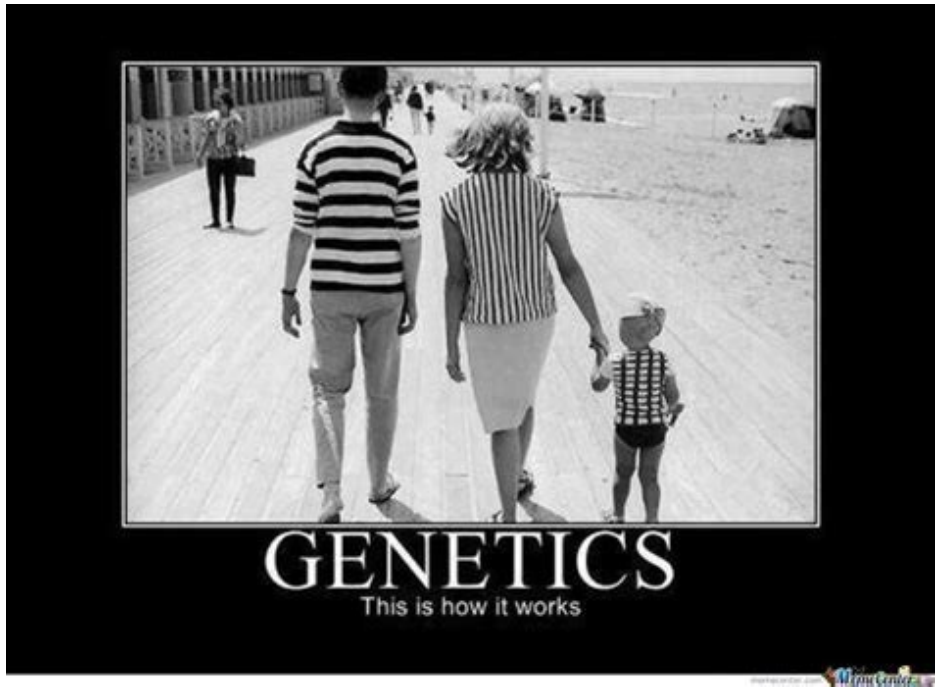
F2

100%

Live poll

Multiple choice

What kind of dominance is this?



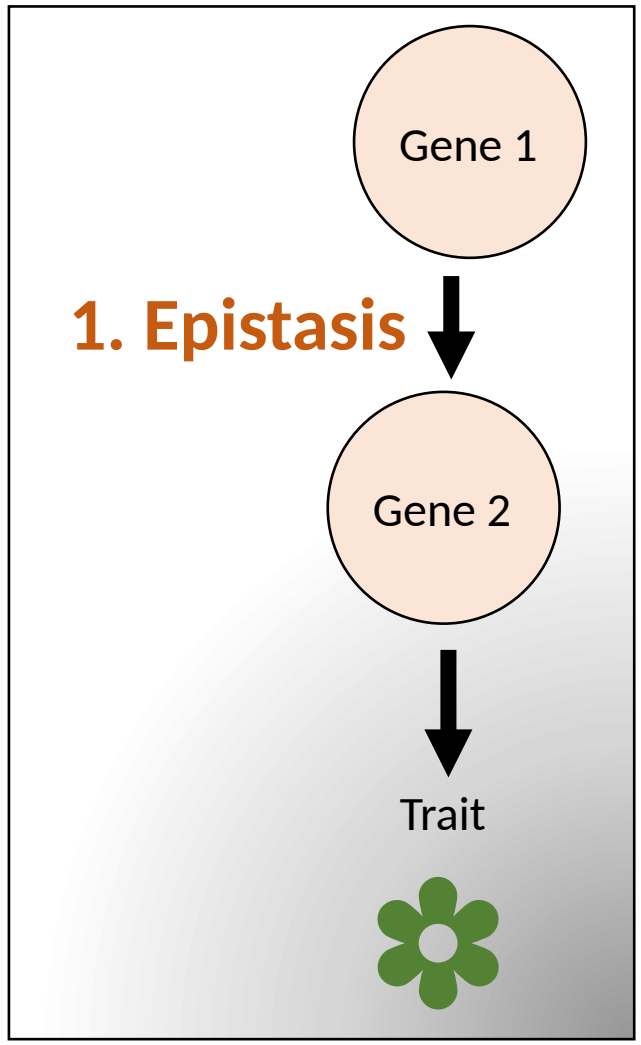
Scan with your phone

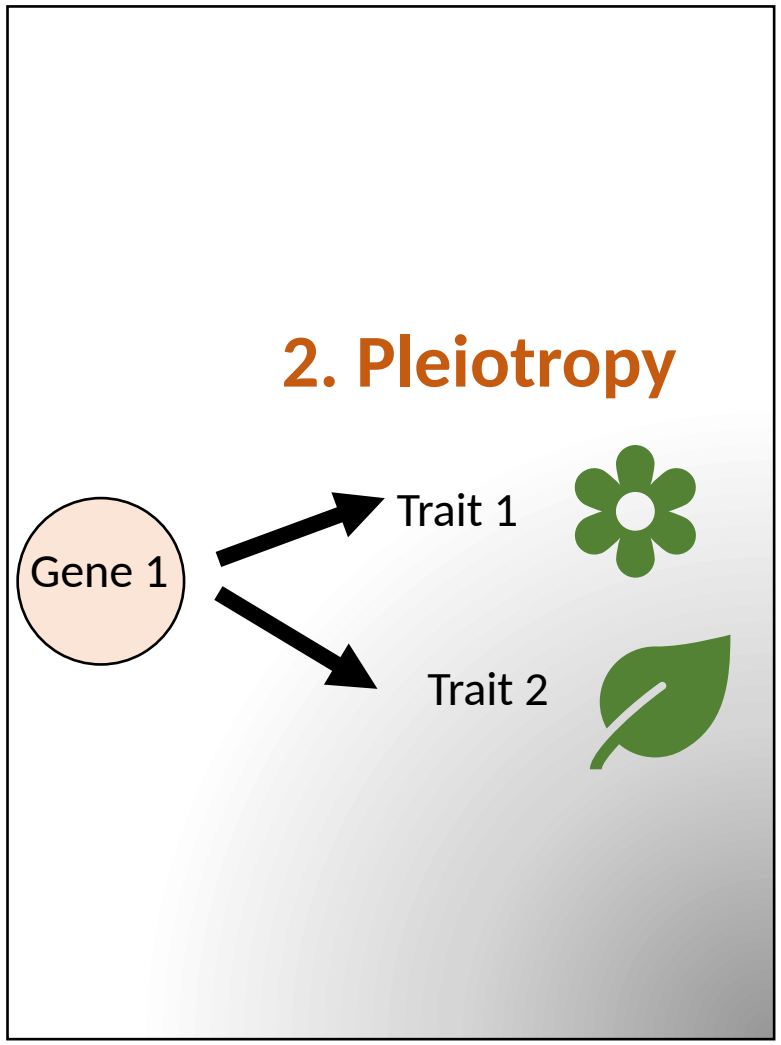
[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-stripes]

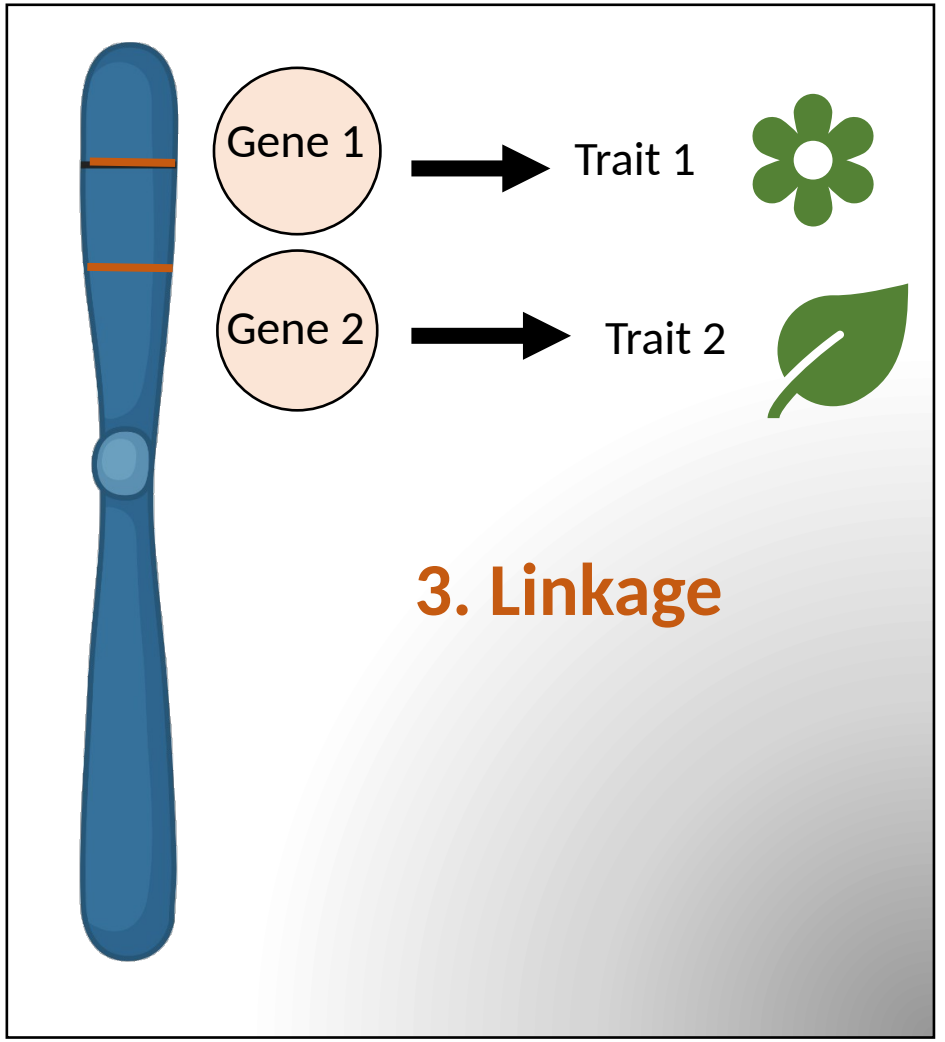
[host screen]

Break ☺

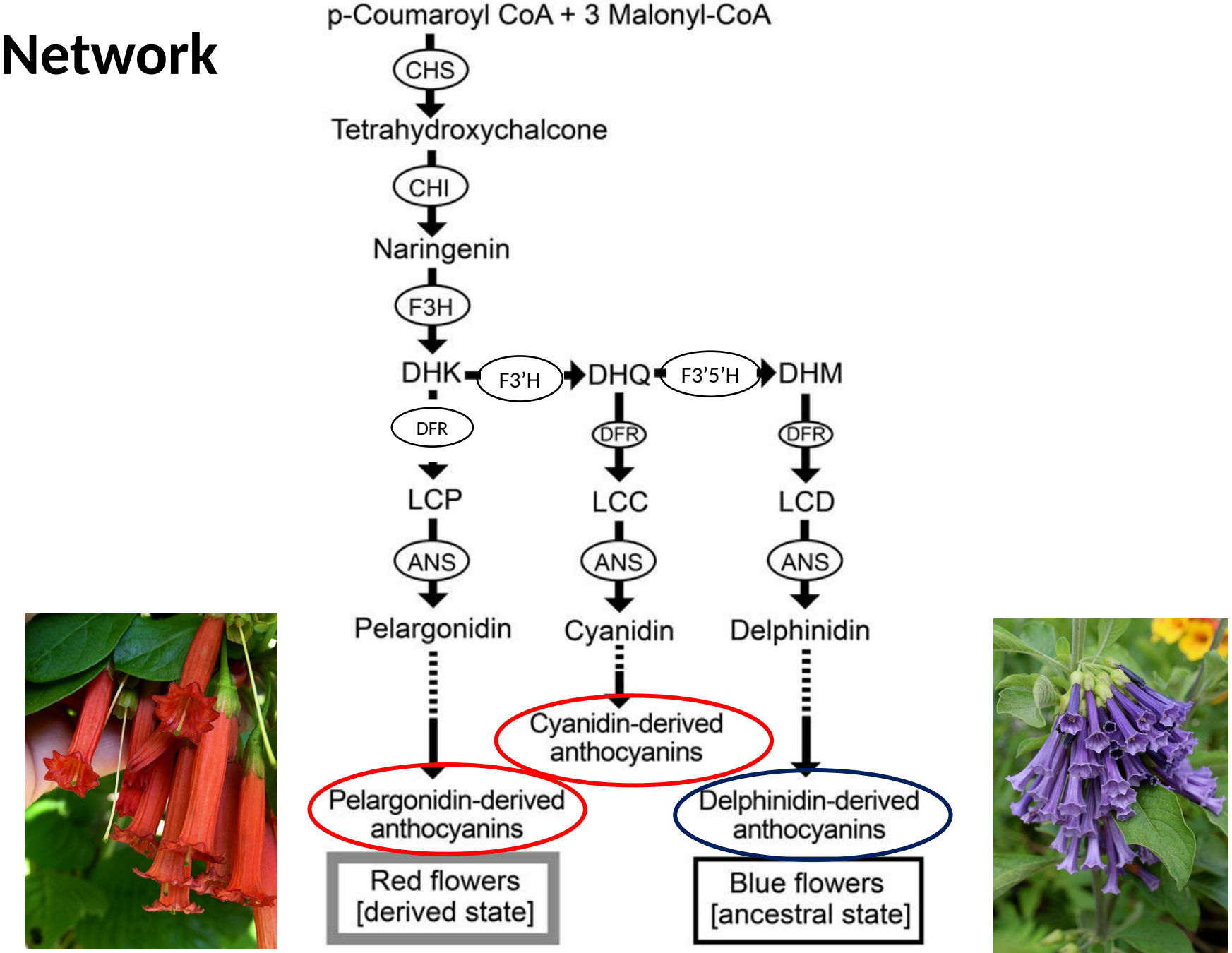
Non-independence between genes and traits



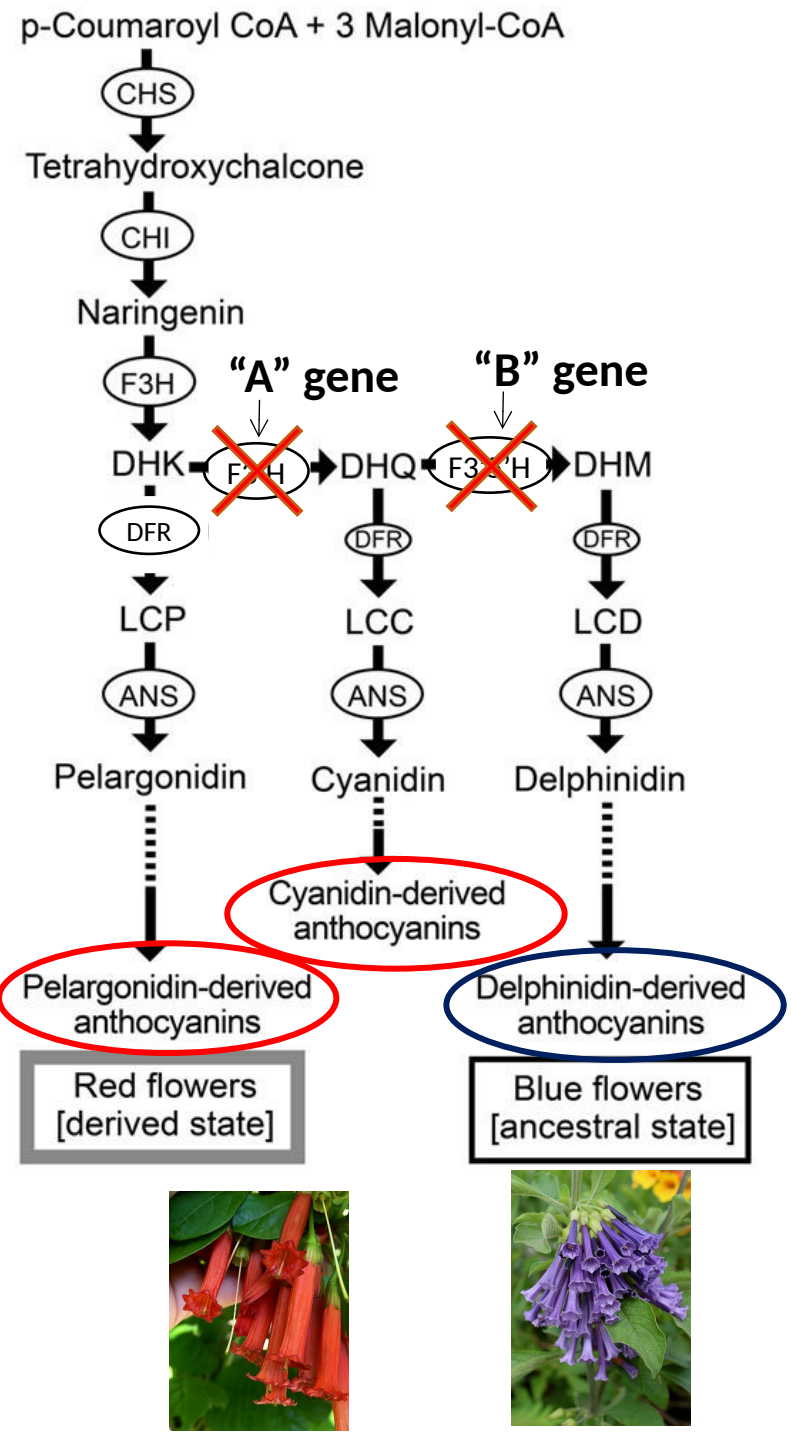




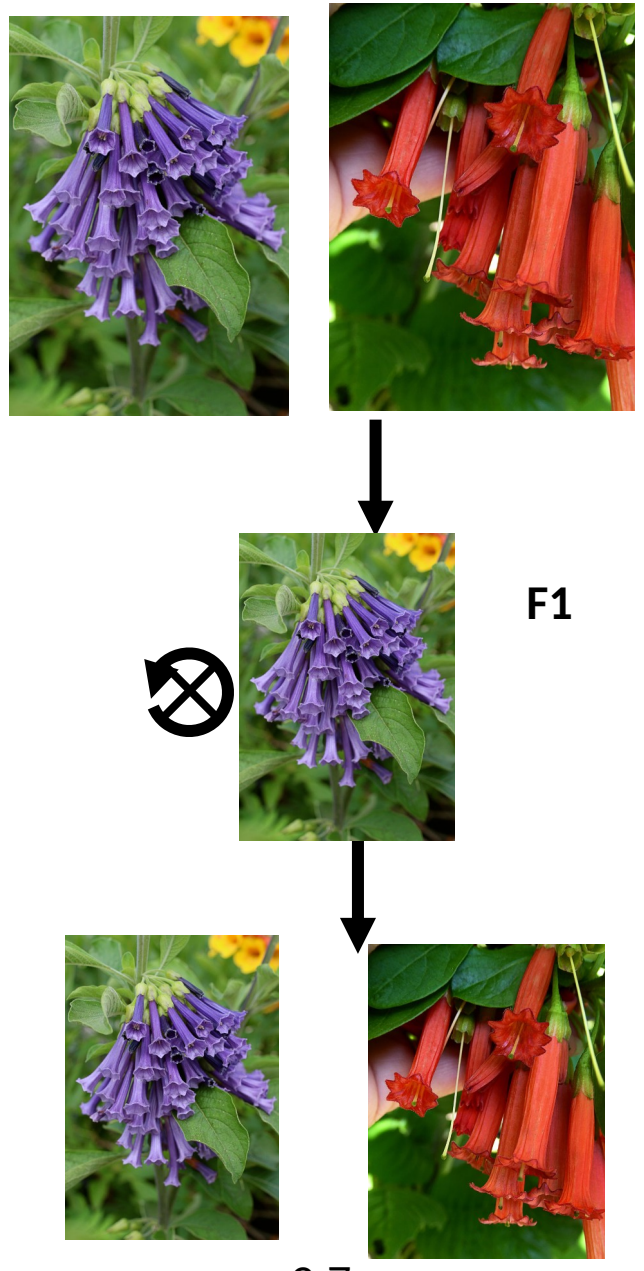
Metabolic Network



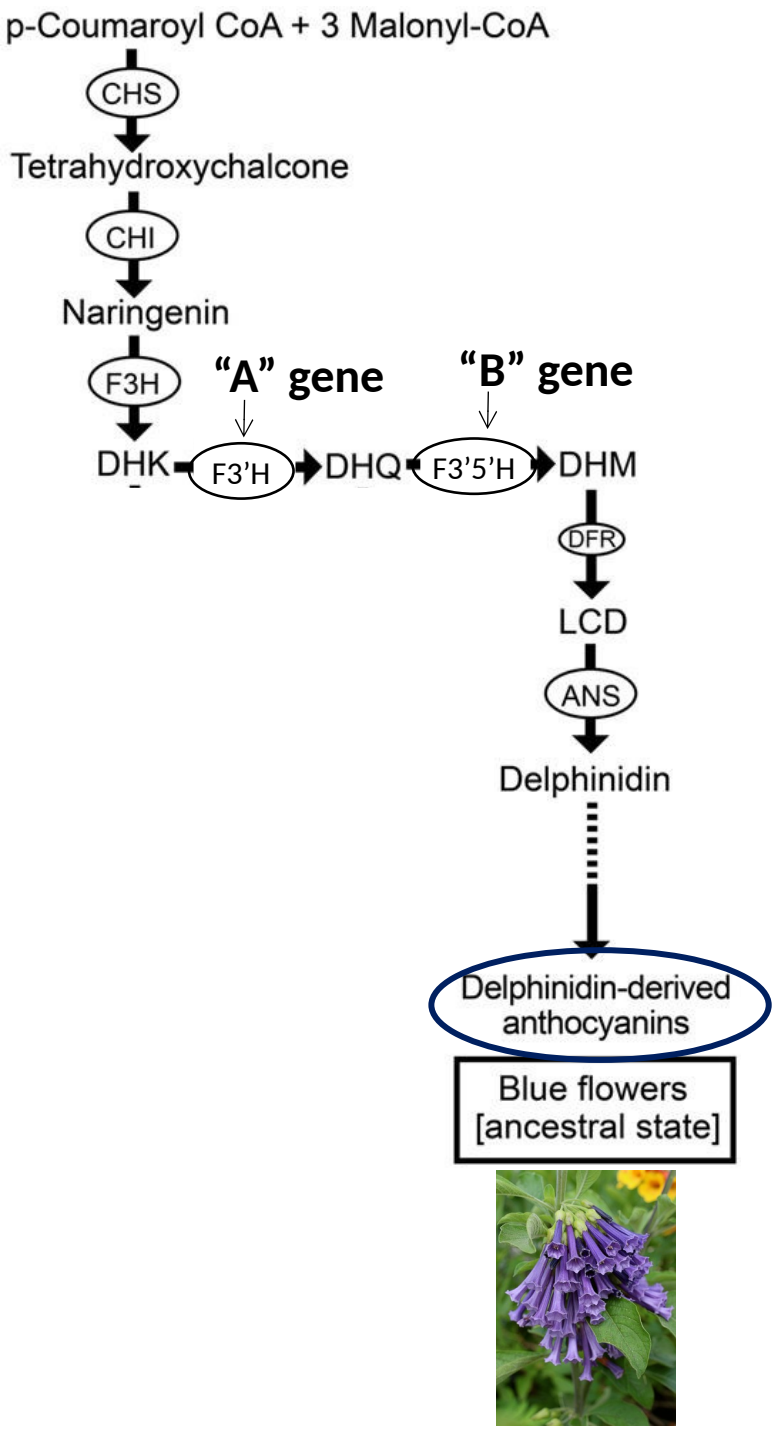
Epistasis



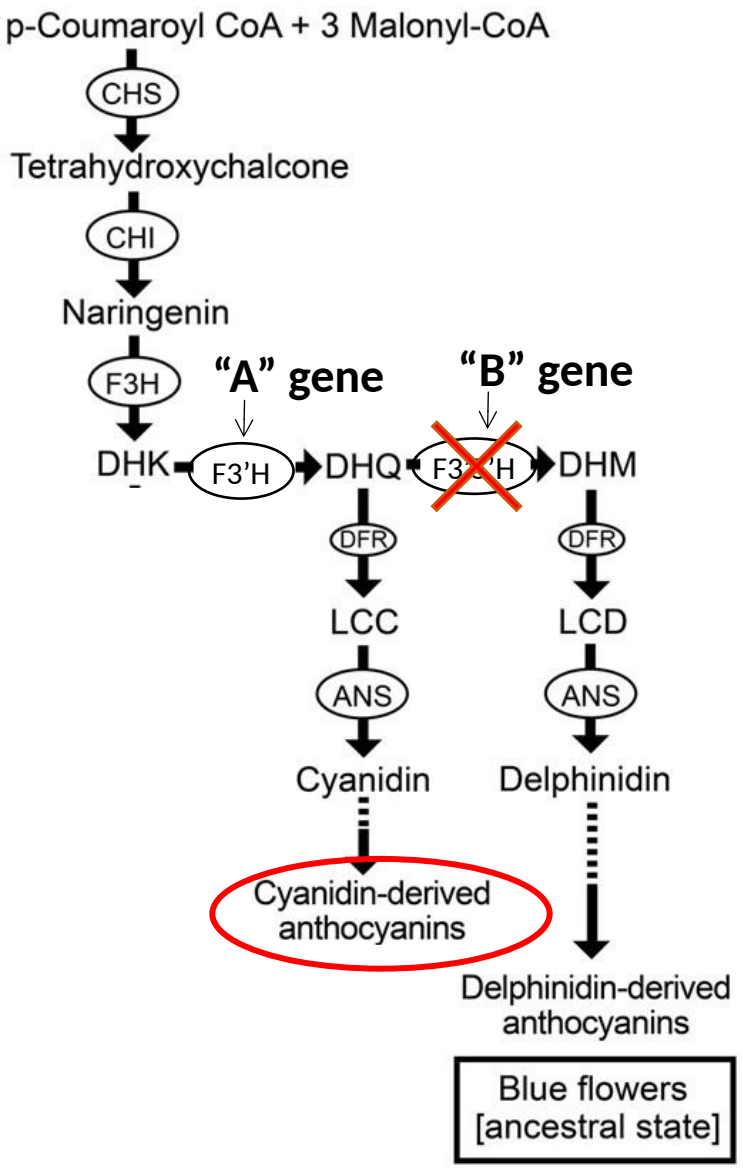
Ipomea



Epistasis

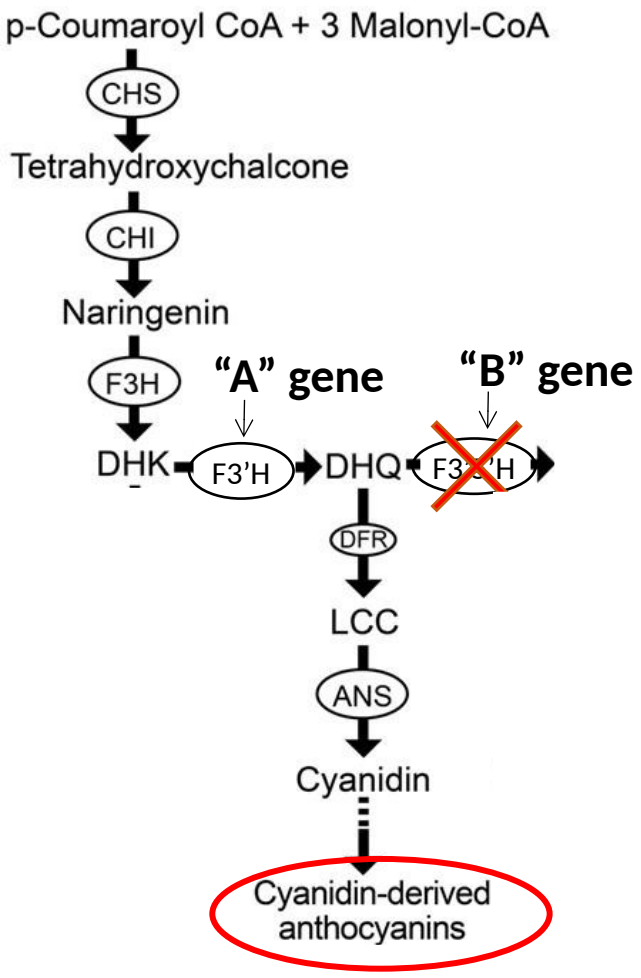

p-Coumaroyl CoA + 3 Malonyl-CoA
CHS
Tetrahydroxychalcone
CHI
Naringenin
F3H
"A" gene
"B" gene
DHK
F3'H
DHQ
F3'5'H
DHM
DFR
LCD
ANS
Delphinidin
Delphinidin-derived anthocyanins
Blue flowers [ancestral state]

Epistasis


p-Coumaroyl CoA + 3 Malonyl-CoA
CHS
Tetrahydroxychalcone
CHI
Naringenin
F3H
"A" gene
"B" gene
DHK
F3'H
DHQ
F3'5'H
DHM
DFR
DFR
LCC
LCD
ANS
ANS
Cyanidin
Delphinidin
Cyanidin-derived anthocyanins
Delphinidin-derived anthocyanins
Blue flowers [ancestral state]

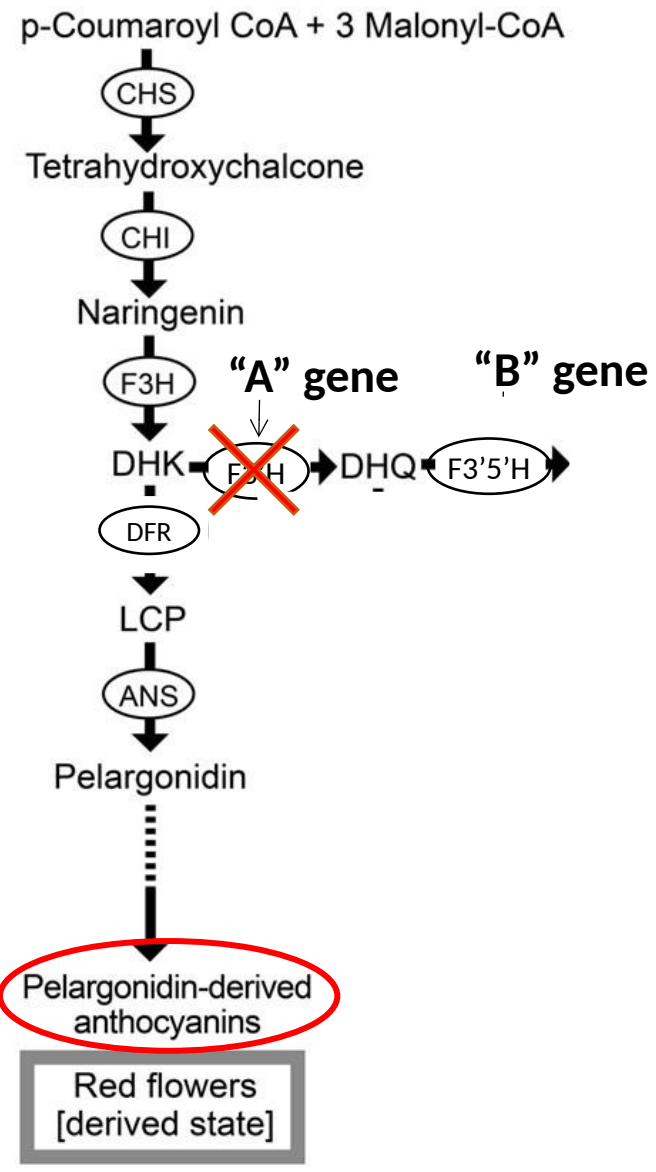


Epistasis


p-Coumaroyl CoA + 3 Malonyl-CoA
CHS
Tetrahydroxychalcone
CHI
Naringenin
F3H
"A" gene
"B" gene
DHK
F3'H
DHQ
DFR
LCC
ANS
Cyanidin
Cyanidin-derived anthocyanins

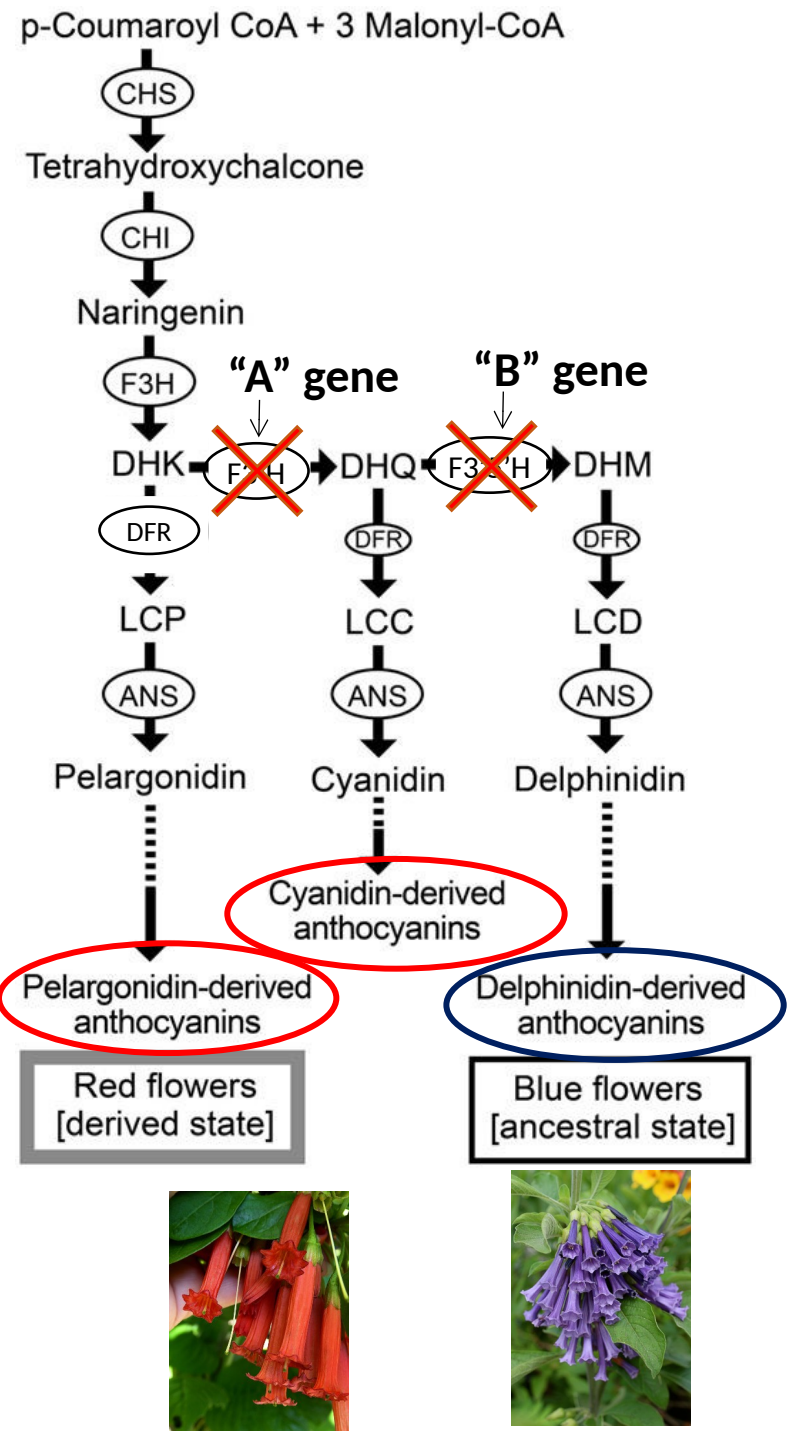


Epistasis


p-Coumaroyl CoA + 3 Malonyl-CoA
CHS
Tetrahydroxychalcone
CHI
Naringenin
F3H
"A" gene
"B" gene
DHK
F3'H
DHQ
F3'5'H
DFR
LCP
ANS
Pelargonidin
Pelargonidin-derived anthocyanins
Red flowers [derived state]



Epistasis



	A,B	A,b	a,B	a,b
A,B				
A,b				
a,B				
a,b				

AABB	AaBB	aaBB
AABb	AaBb	aaBb
AAbb	Aabb	aabb

aa OR bb causes red phenotype

Ipomea



AABB



aabb



F1

AaBb

A-B-

aaB- or A-bb or aabb





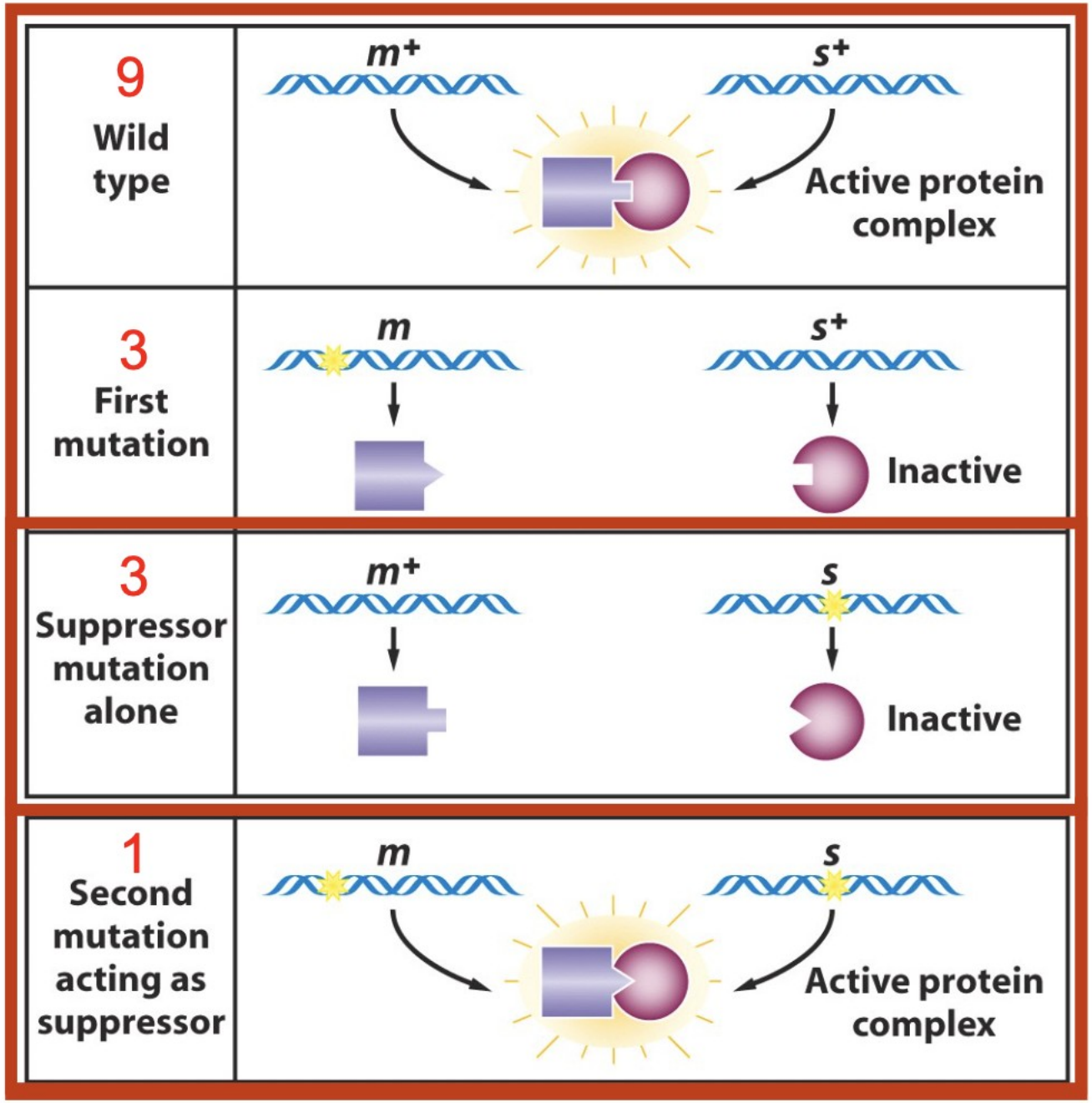
F2

9:7

Suppressor mutations F_2 ratio = 10:6



Green revolution wheat



Synthetic lethality: F2 ratio = 3:2 (9:6)

	A+ B+	A- B+	A+ B-	A- B-
A+ B+	😎	😎	😎	😎
A- B+	😎	☹	😎	☹
A+ B-	😎	😎	☹	☹
A- B-	😎	☹	☹	☠

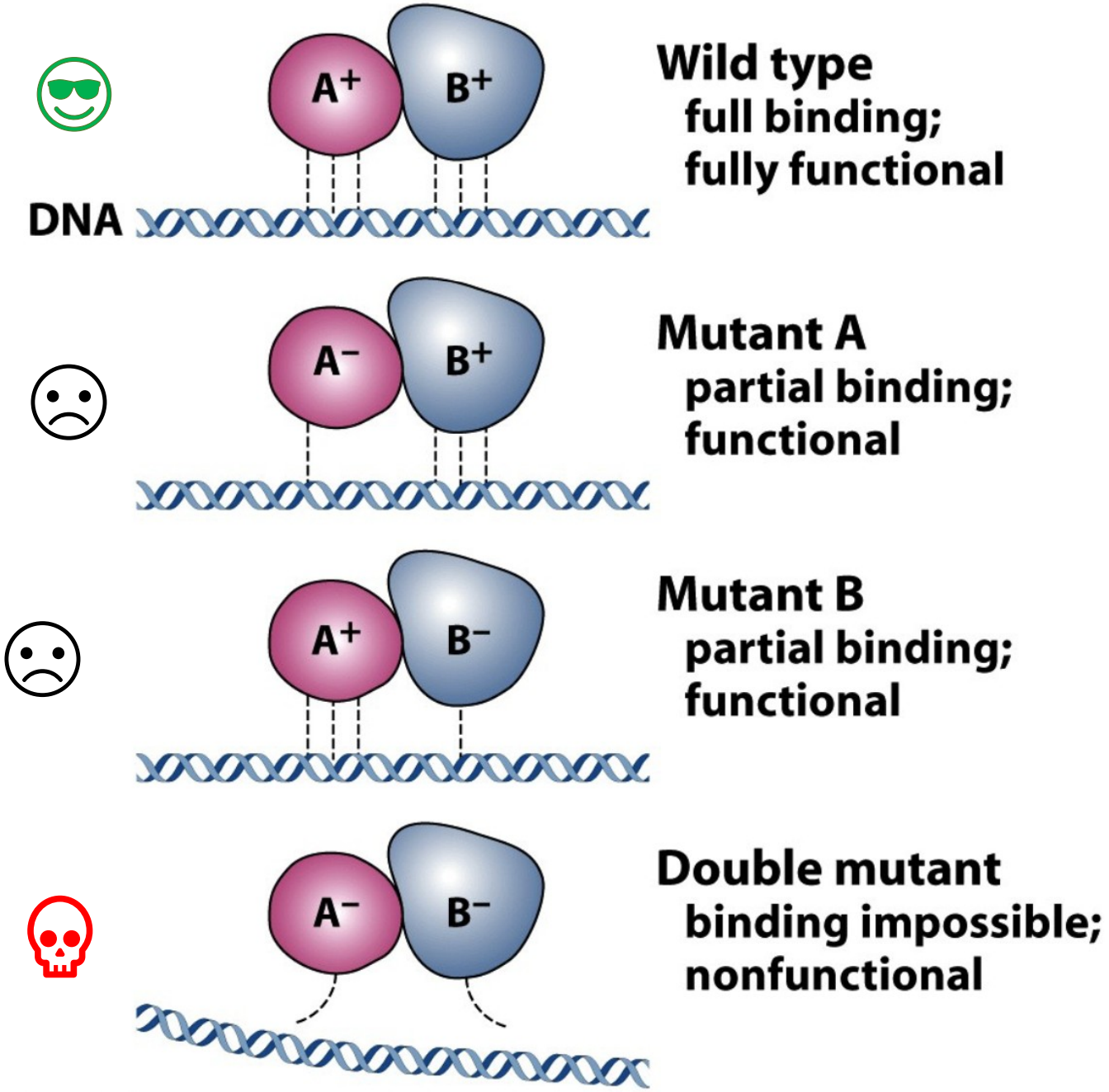
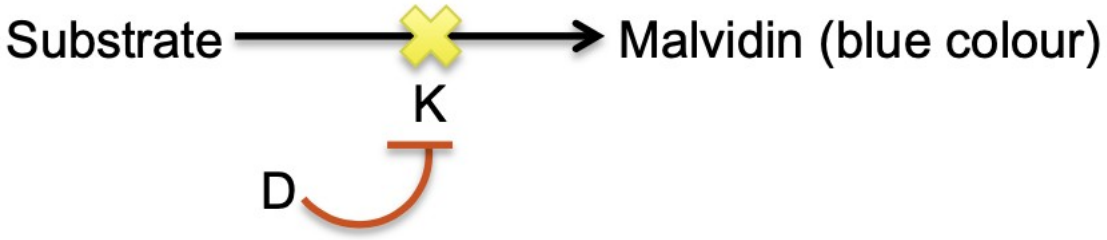


Figure 6-23
Introduction to Genetic Analysis, Ninth Edition
© 2008 W. H. Freeman and Company





Primula



- **Functional K allele needed to make blue pigment**
- **Functional D allele prevents K from acting**

What is the expected fraction of blue offspring from a dihybrid cross:

KkDd x KkDd

Hints: make dihybrid Punnet square

Live poll

Multiple choice

Primula, KkDd × KkDd: what fraction of the offspring are blue?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-primula

host screen

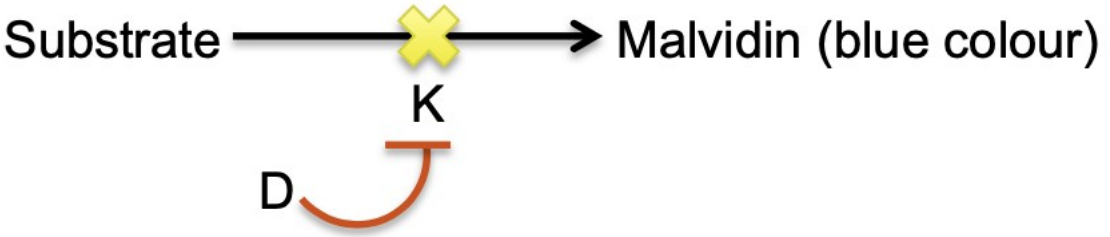
Dominant epistasis

		Female Gametes			
Male Gametes		*KD*	*Kd*	*kD*	*kd*
	KD	*KKDD*	*KKDd*	*KkDD*	*KkDd*
	Kd	*KKDd*	*KKdd*	*KkDd*	*Kkdd*
	kD	*KkDD*	*KkDd*	*kkDD*	*kkDd*
	kd	*KkDd*	*Kkdd*	*kkDd*	*kkdd*

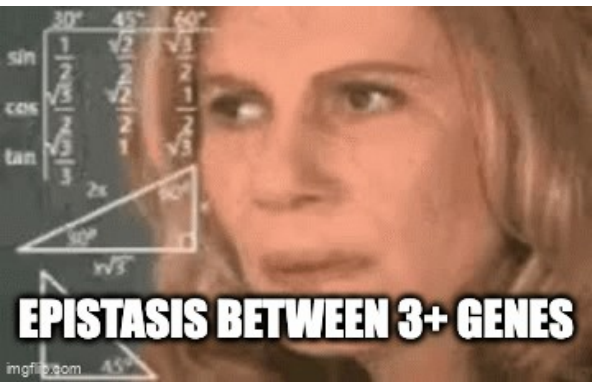




Primula

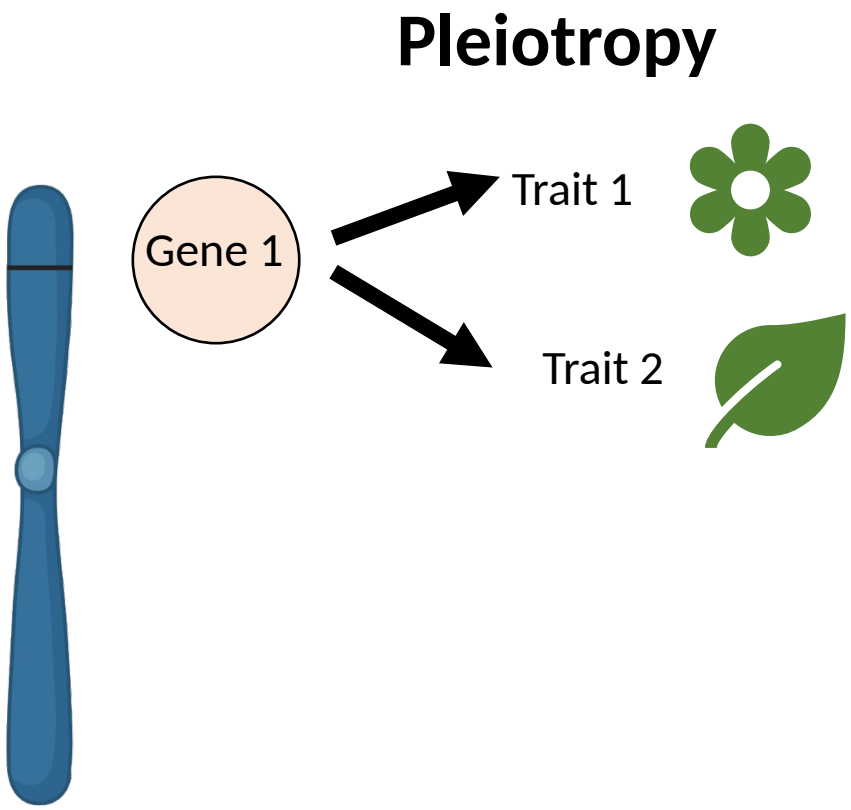


- **Functional K allele needed to make blue pigment**
- **Functional D allele prevents K from acting**



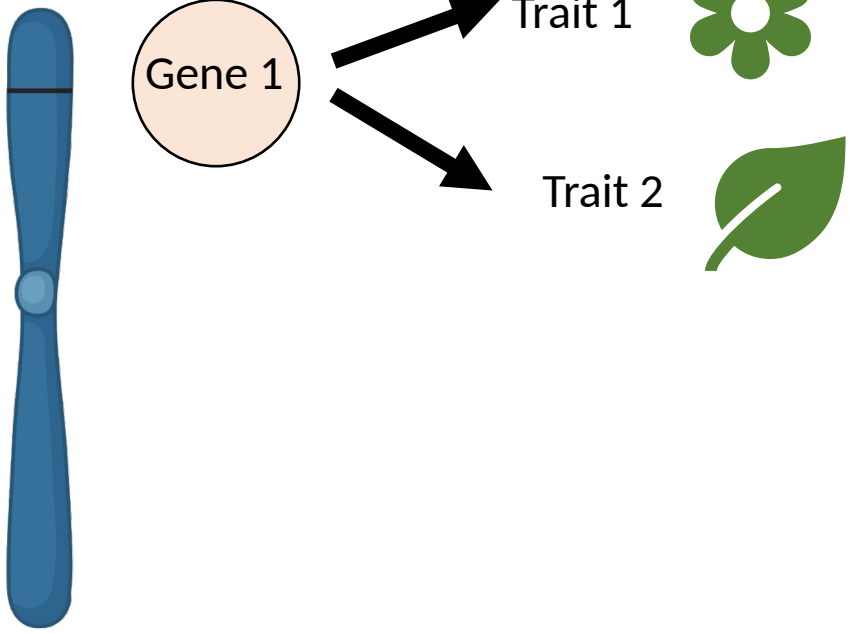
vernalization
circadian clock
autonomous
photoperiod
ambient temperature
Gibberellin (GA_4)
PAF1-like complex
CDC73/PHP
VIP2/ELF7
VIP4,5
VIP6/ELF8
SWR1/SRCAP-like complex
PIE1
ARP6/SUF3
SEF
VRN2 complex
VRN2
FIE
CLF/SWN
MSI1
VIN3
VRN1
ATX1,2,7
EFS
EMF2
ELF5
RNA binding
FCA
FPA
FLK
RNA processing
FY
Chromatin modification
LD
FLD
FVE/MSI4
Histone modification
SWP1/LDL1
REF6
CZS
-responsive miRNAs
163
169
398
399
PHO2
FRI
FRL1,2
SUF4
FES
GRP7
FLC
MAF2-5
PEP
BBX24
FLM
SPA1
HUA2
LHP1/ TFL2
HDA9
EBI
ZTL
FKF1
LKP2
GI
CDF1-3,5
PIF4
VOZ1,2
HOS1
phyB
phyD
phyE
phyA
PFT1
CBF/NF-Y
HAP2,3,5
ICE1
DET1
PRR3
RVE8
PRR9
PRR7
PRR5
LHY/CCA1
CHE
TOC1
ELF3 ELF4
LUX
DDB1
CUL4
COP1
SPAs
CRY1
CRY2
CIB1
TEM1,2
AGL19
miR824
AGL16
SVP
BBX19
CO
AS1
FBH1-4
EDL3
OBF4
DNF
FT
AGL17
FD
TSF
AGL24
SOC1
TOE1,2 AP2
SNZ, SMZ
miRNA156
miRNA172
age
SPL9,10
GA20ox
GA3ox
SPL9
GID1
SLY1
DELLA
RGA
GAI
RGL1-3
PIFs
GNC/ GNL
FPF1
TFL1
FUL
CAL
SEPs
LFY
AP1
SPL3-5
TPS1
Activation and/or stabilization
Inhibition and/or degradation
Genetic and/or physical interaction
Indirect interaction
Current Opinion in Biotechnology

Pleiotropy: one gene multiple traits



Pleiotropy: one gene multiple traits

Pleiotropy



Rht in wheat

- **Tall**
- **Few tillers**



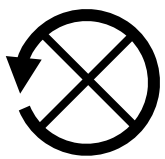
X



- **Short**
- **Many tillers**

- **Tall**
- **Few tillers**







- **Tall**
- **Few tillers**

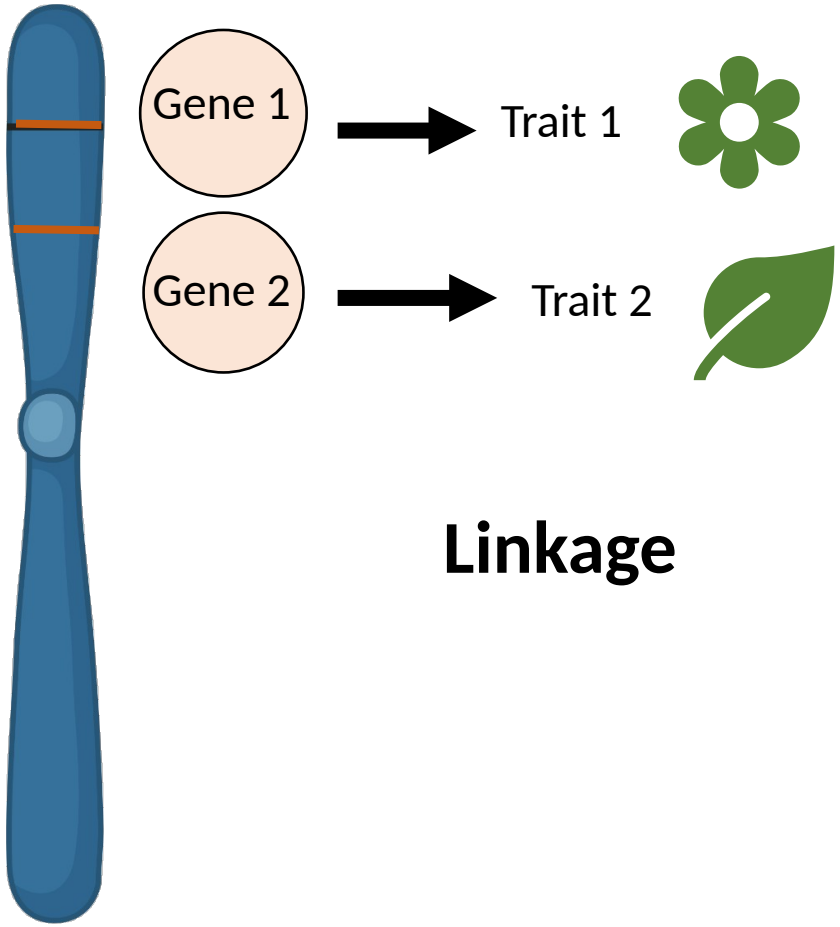
75%



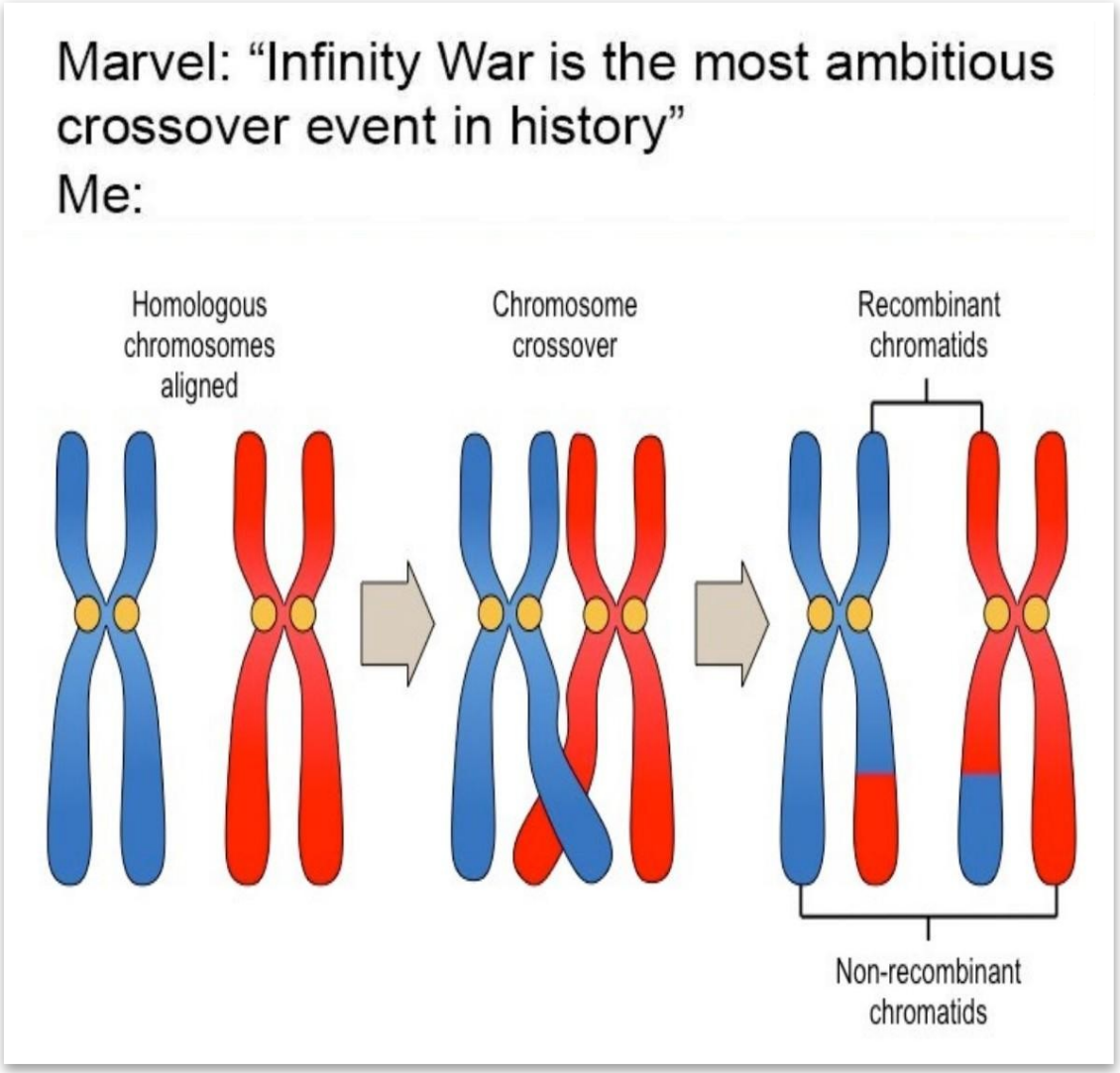
- **Short**
- **Many tillers**

25%

Linkage: two genes on the same chromosome



Linkage

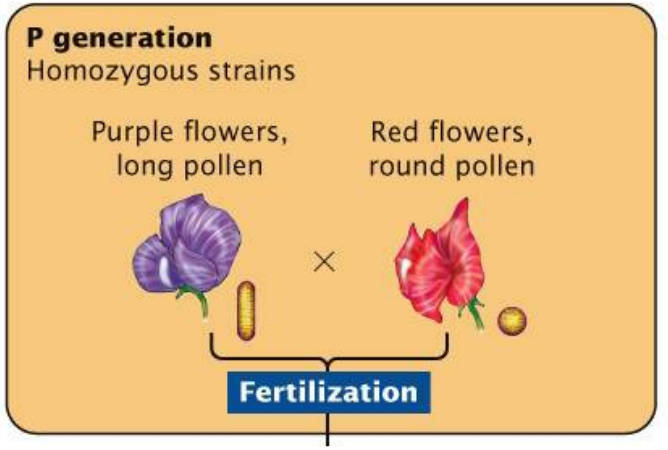






Edith Rebecca Saunders,
Reginald Punnet,
William Bateson

Cross two strains homozygous for two different traits.

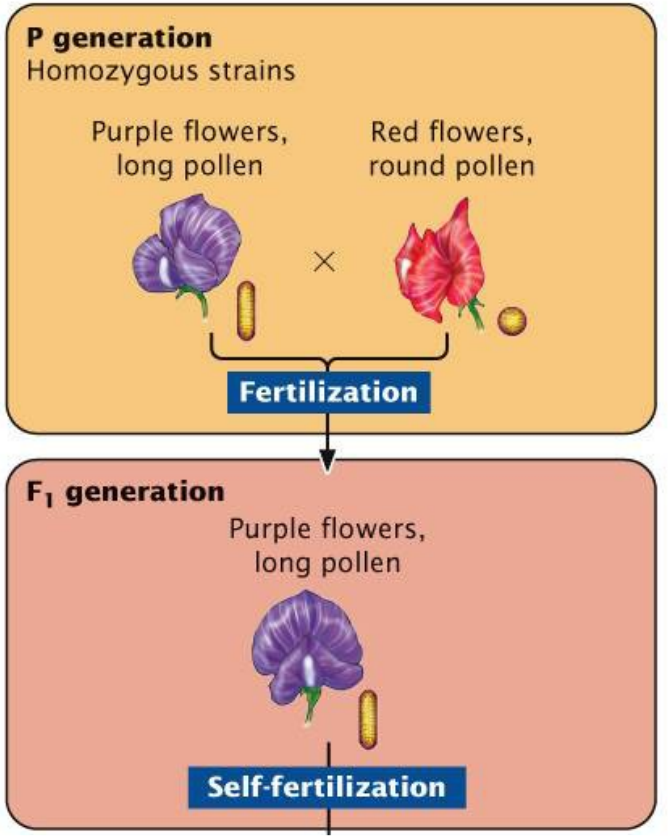






Edith Rebecca Saunders,
Reginald Punnet,
William Bateson

Cross two strains homozygous for two different traits.

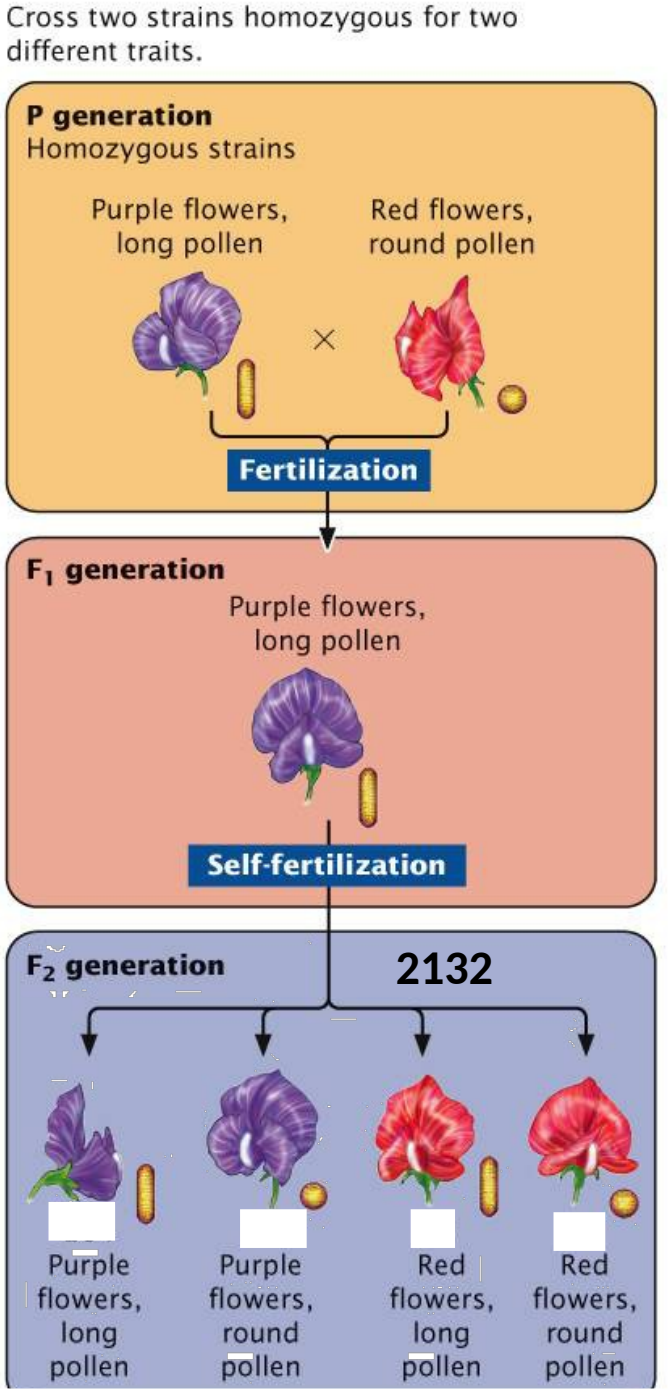






Edith Rebecca Saunders,
Reginald Punnet,
William Bateson

Phenotype	Expected
Purple, long	
Purple, round	
Red, long	
Red, round	
Total	2132

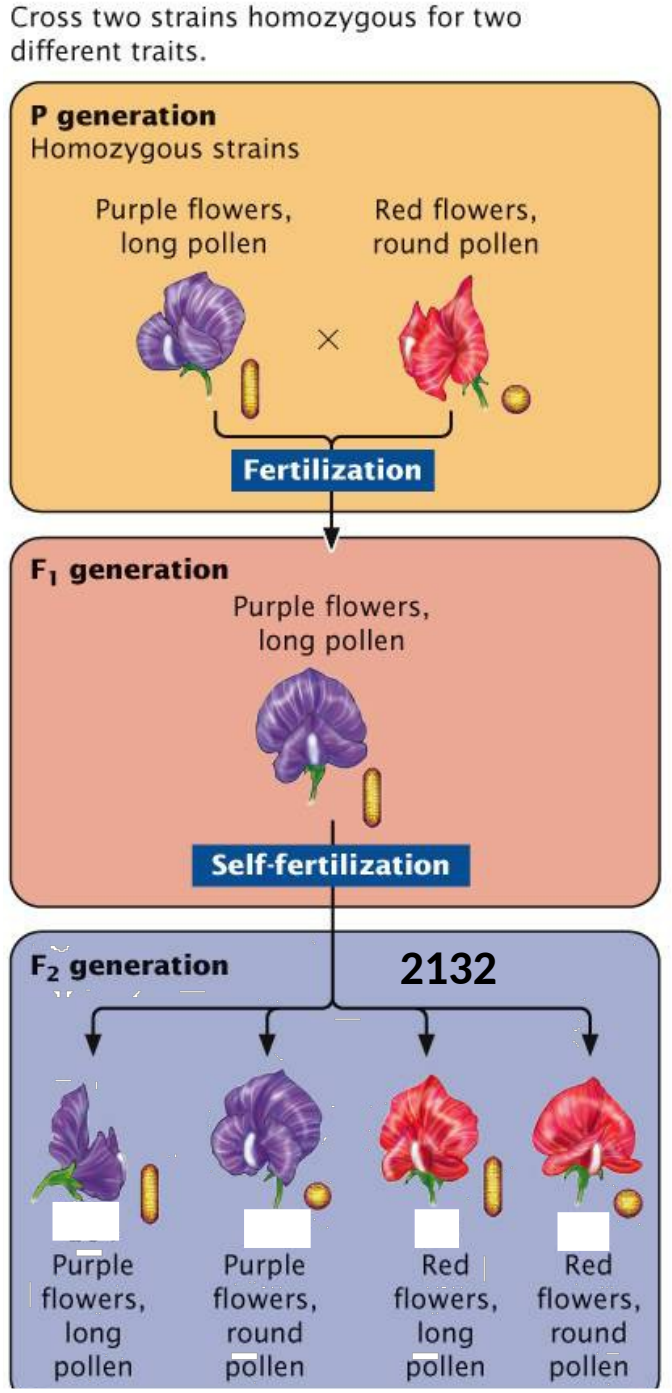






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Reginald Punnet,
William Bateson

Phenotype	Expected
Purple, long	1199
Purple, round	400
Red, long	400
Red, round	133
Total	2132

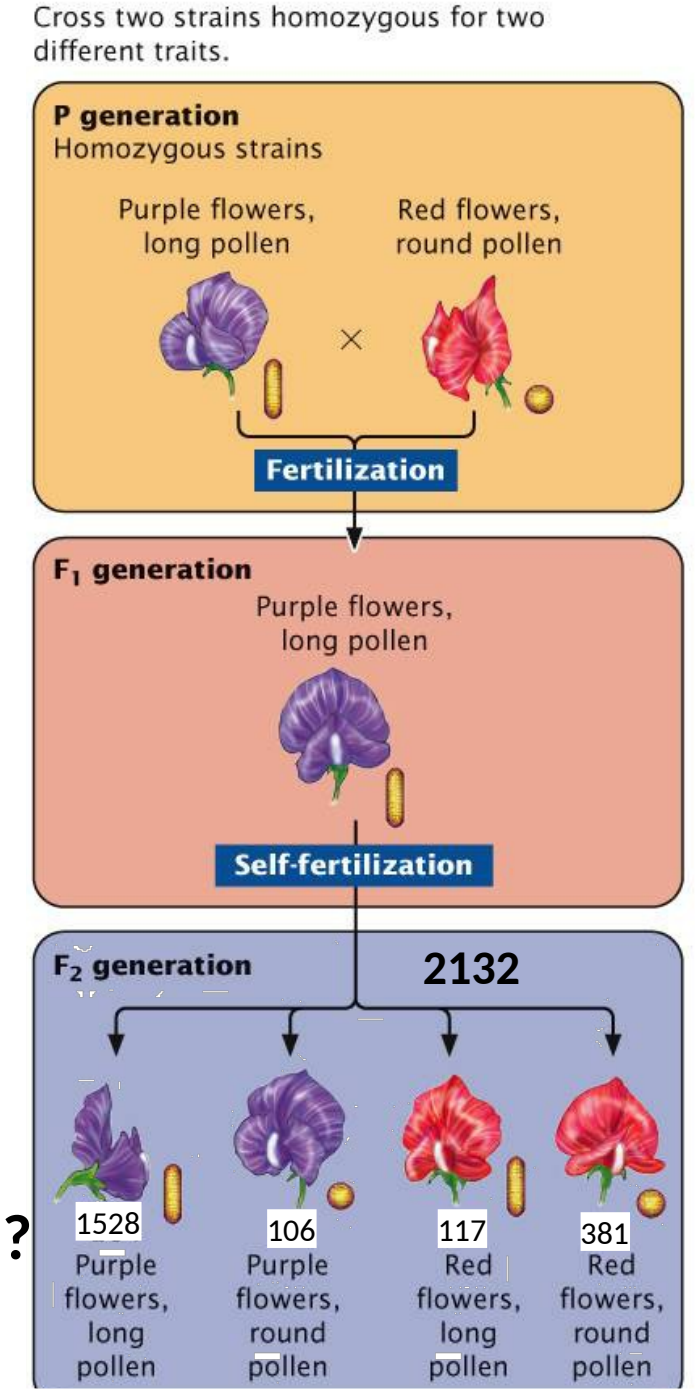






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Reginald Punnet,
William Bateson

Phenotype	Expected	Observed
Purple, long	1199	1528
Purple, round	400	106
Red, long	400	117
Red, round	133	381
Total	2132	2132



Which phenotype combinations fewer observed than expected?

Which phenotype combinations are ”recombinants”?



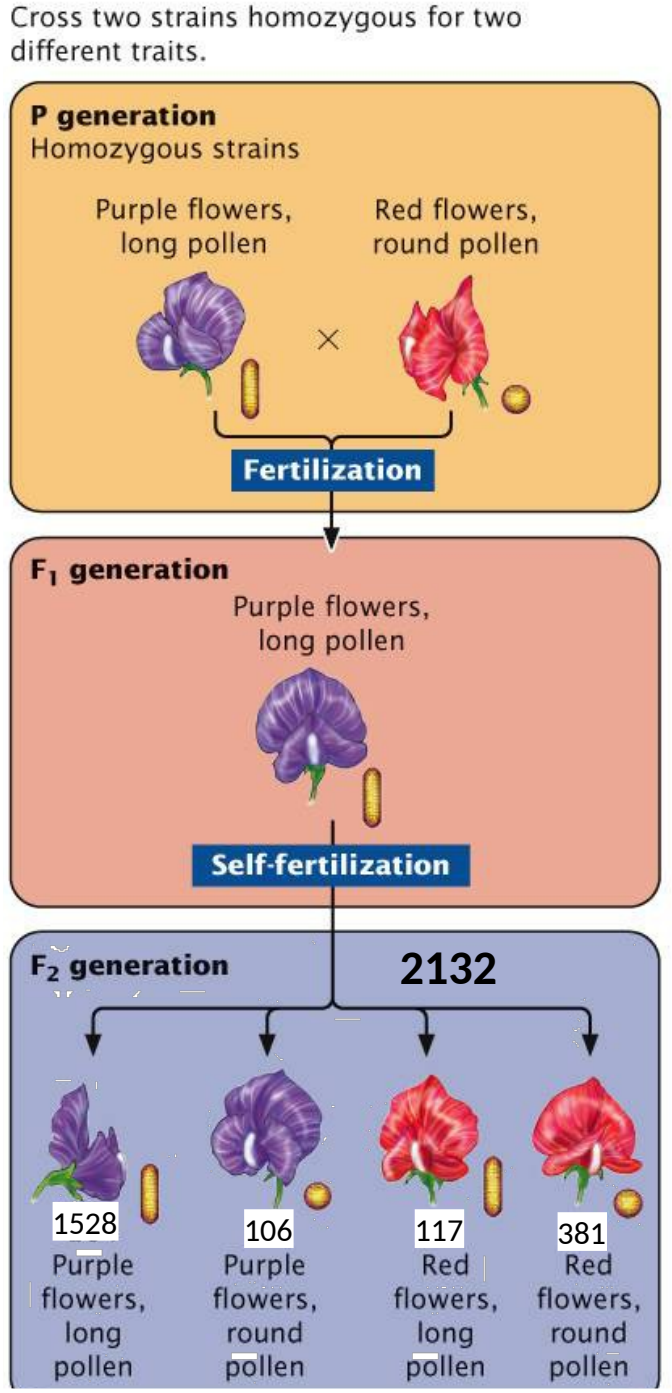


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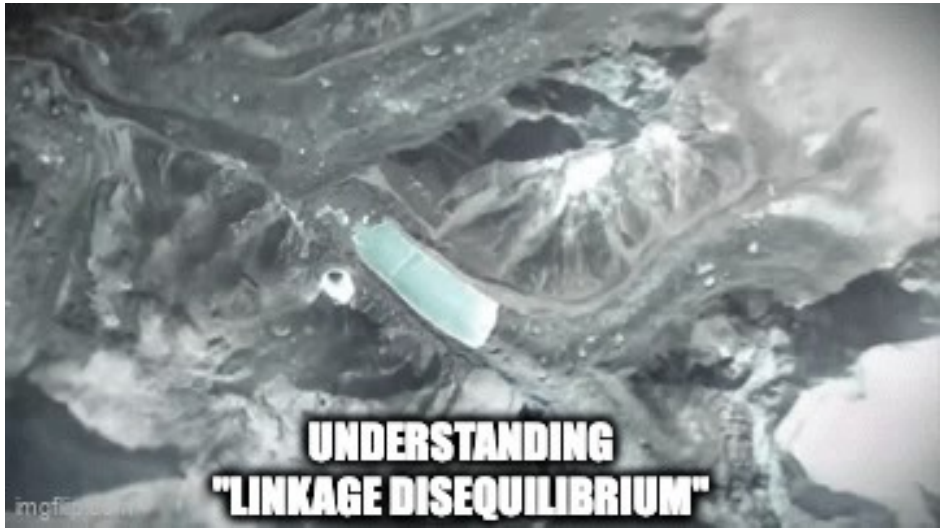
Alleles between genes “mix” as expected… they are in….
”Equilibrium”

Alleles between genes do not “mix” as expected… they are in….
Linkage ”DISequilibrium”

Phenotype	Expected	Observed
Purple, long	1199	1528
Purple, round	400	106
Red, long	400	117
Red, round	133	381
Total	2132	2132

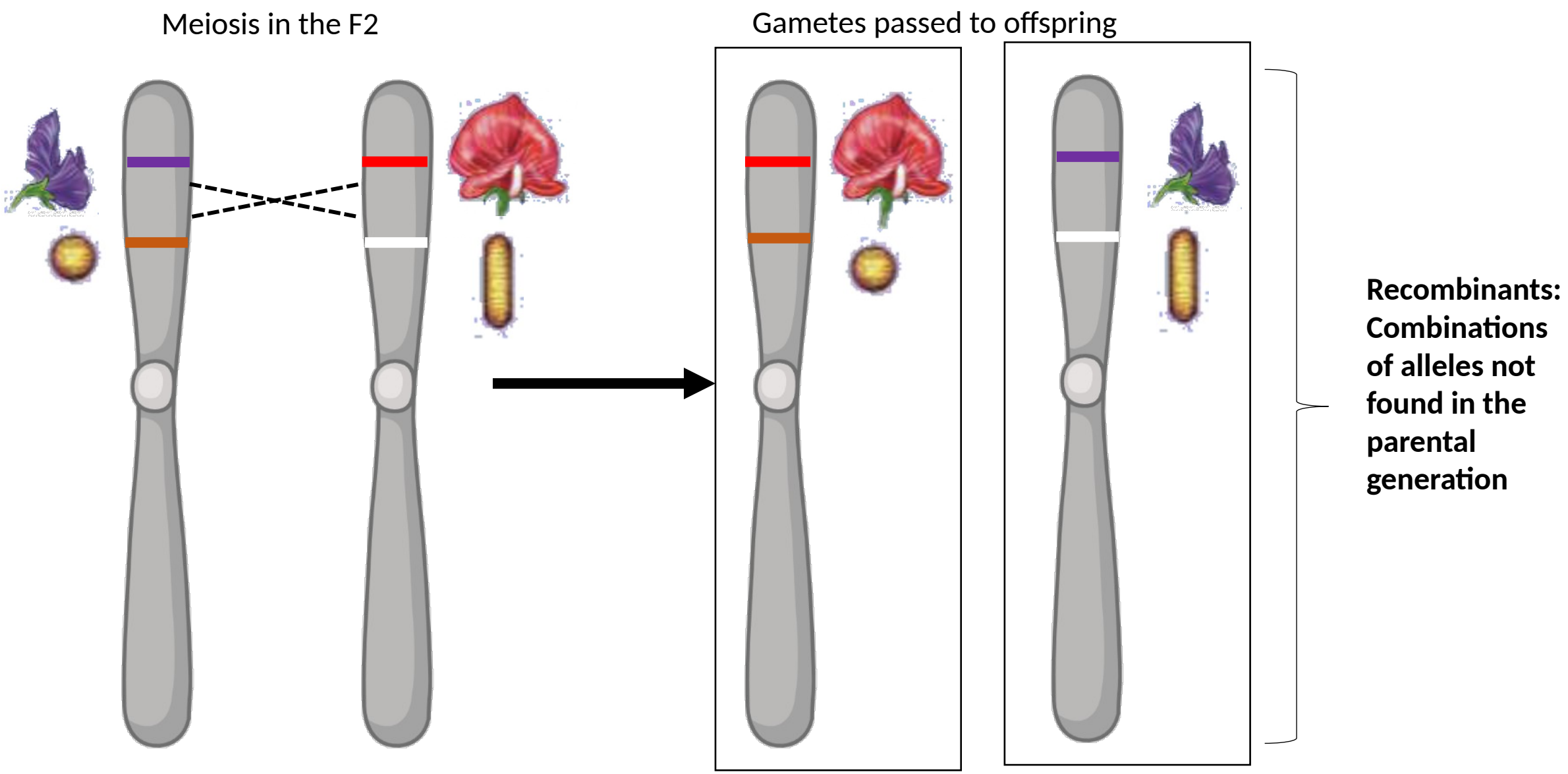


Genes close to each other on the same chromosome show **greater linkage disequilibrium** (LD) and **less recombination**





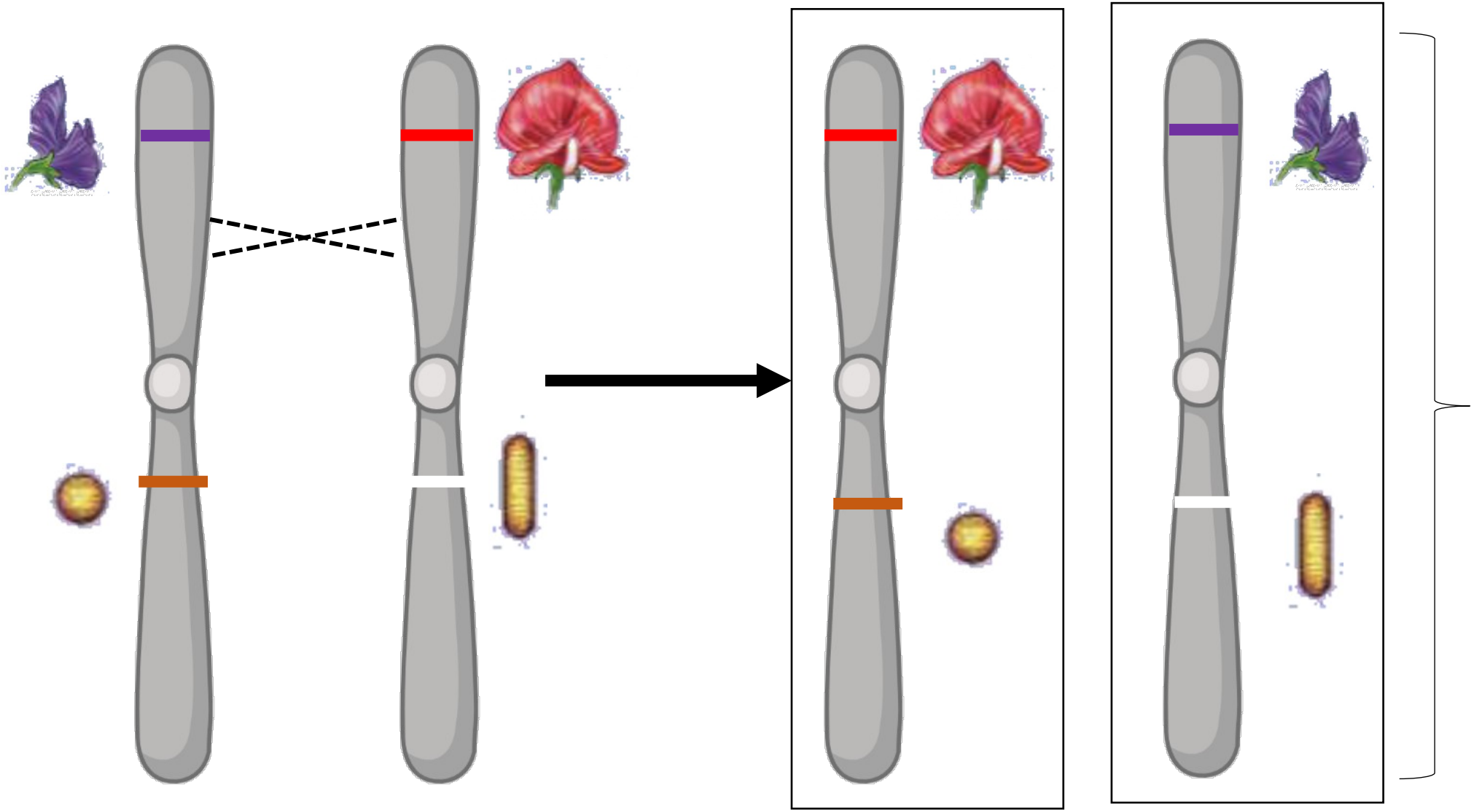




Recombination

Meiosis in the F2

Gametes passed to offspring

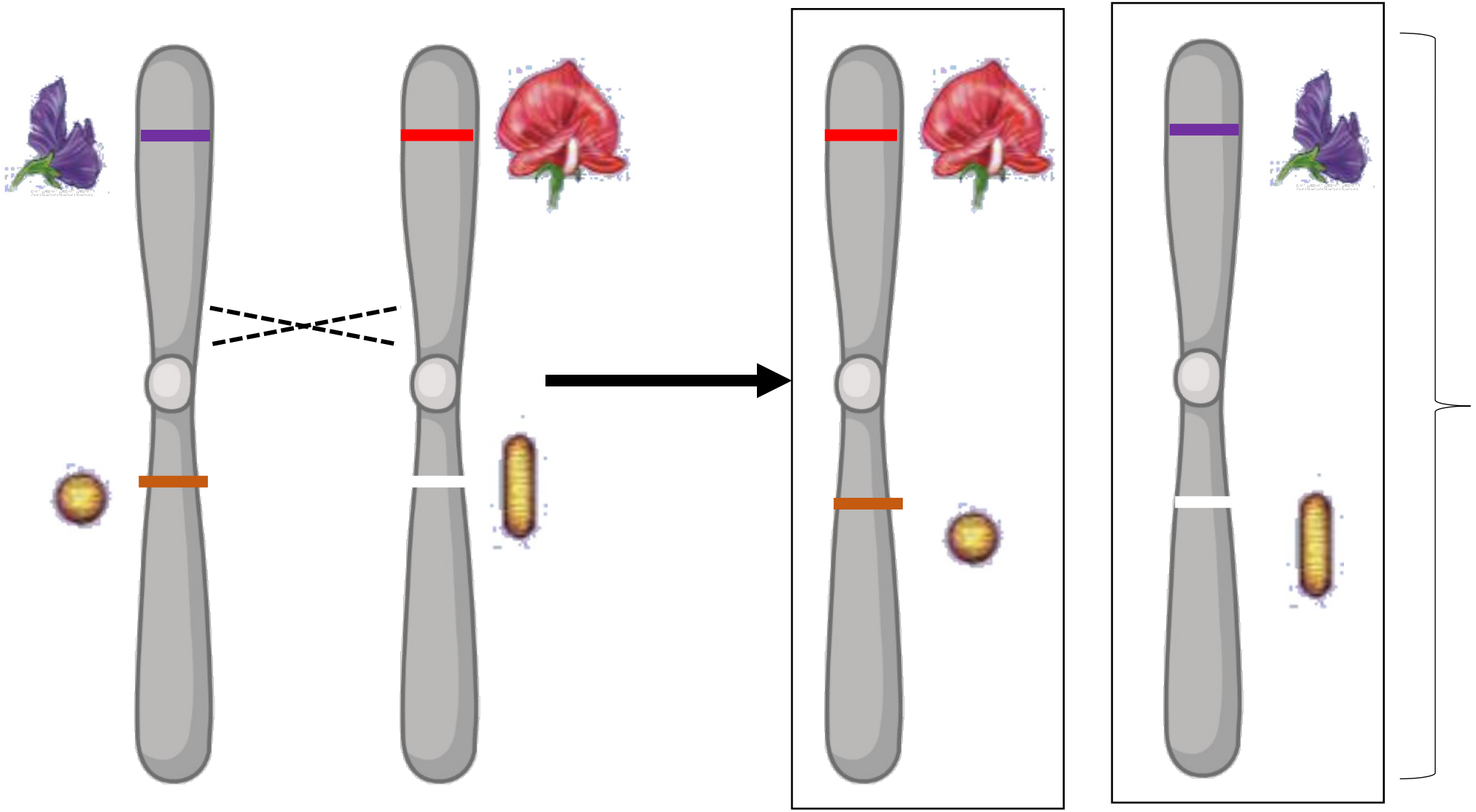


Recombinants: Combinations of alleles not found in the parental generation

Further the genes, the more opportunities for **Recombination**

Meiosis in the F2

Gametes passed to offspring

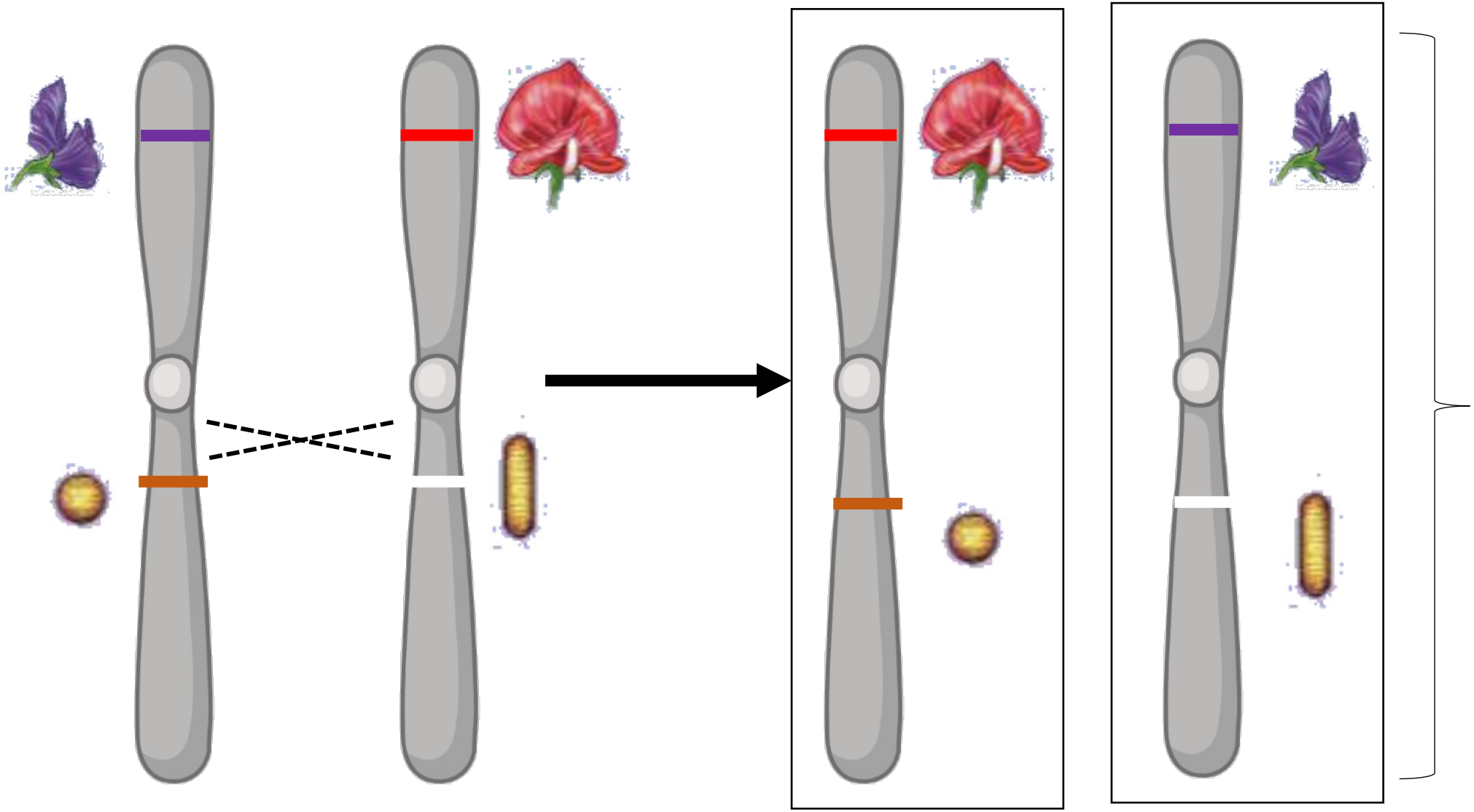


Recombinants: Combinations of alleles not found in the parental generation

Further the genes, the more opportunities for **Recombination**

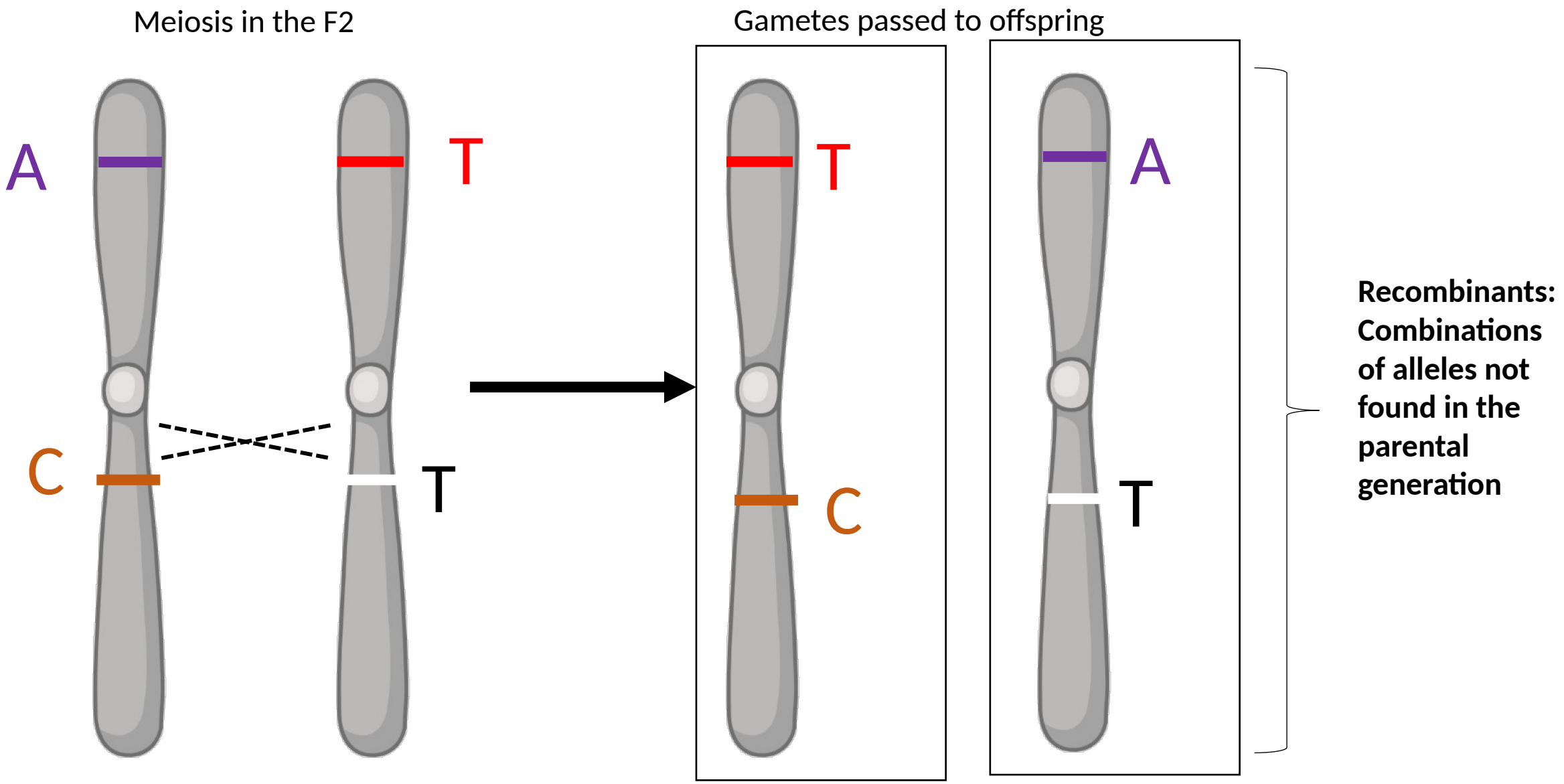
Meiosis in the F2

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Recombinants: Combinations of alleles not found in the parental generation

Further the genes, the more opportunities for **Recombination**



Further the genes, the more opportunities for **Recombination**

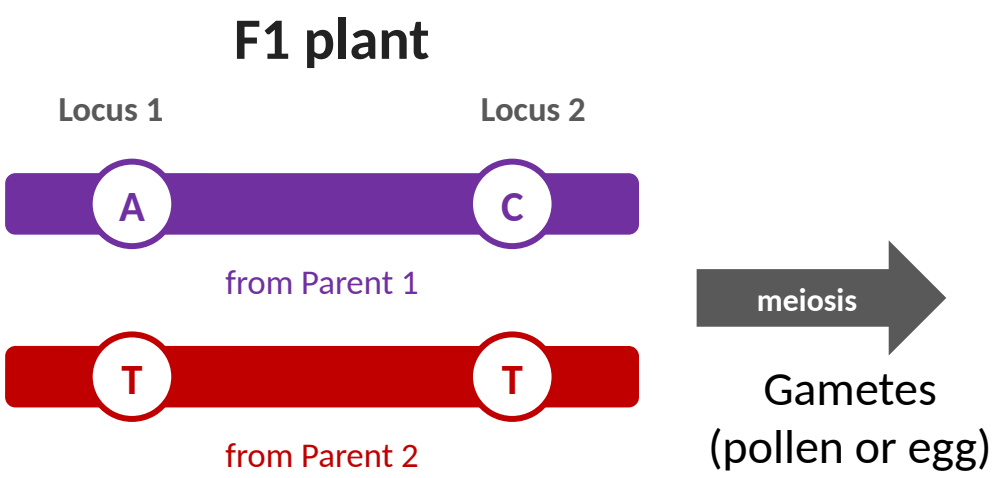
Recombination: seeing it in DNA

Where it happens • how we count it • how it builds a map

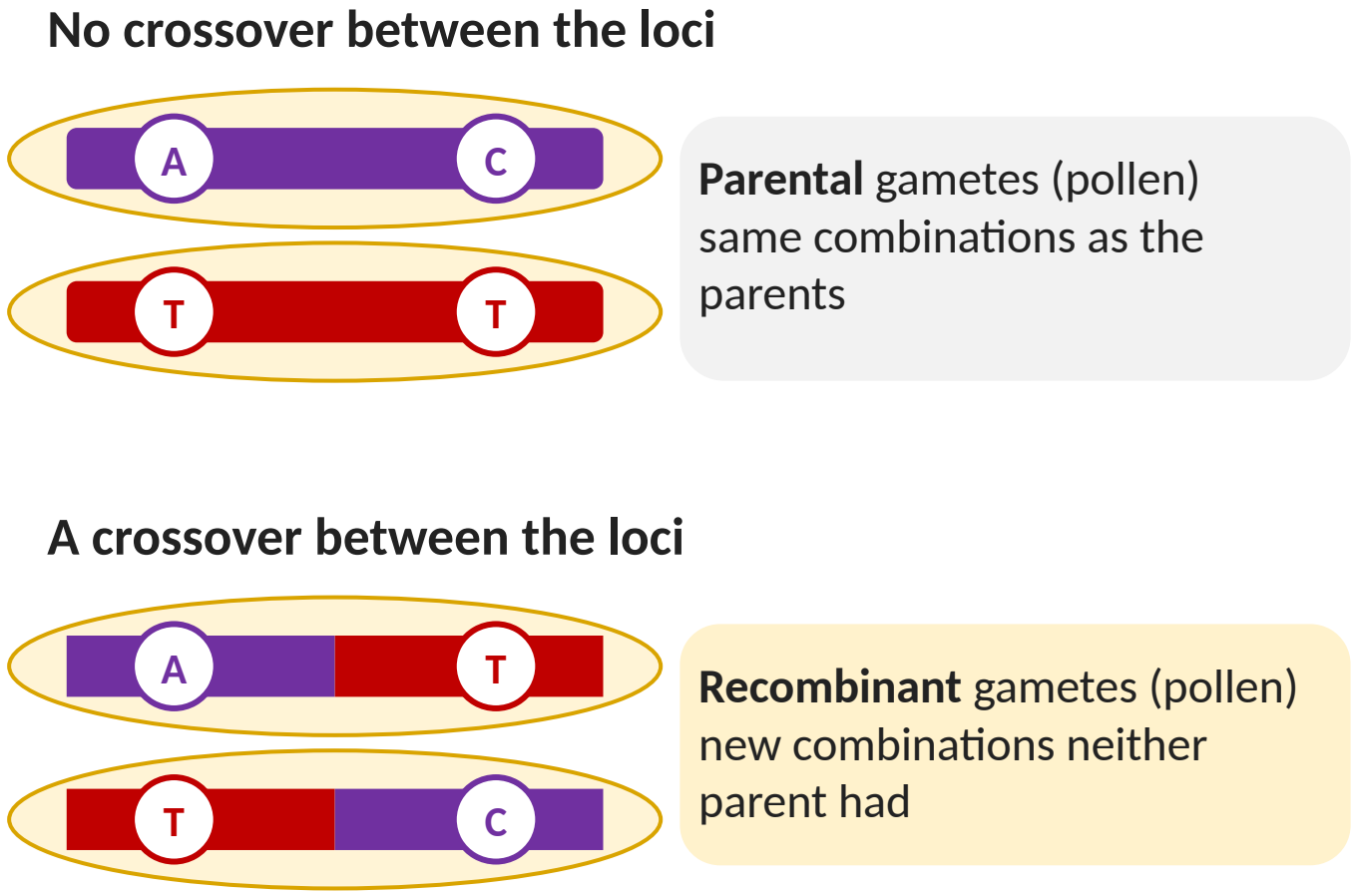


Crossovers happen during meiosis in the F1

The F1 carries one chromosome from each parent

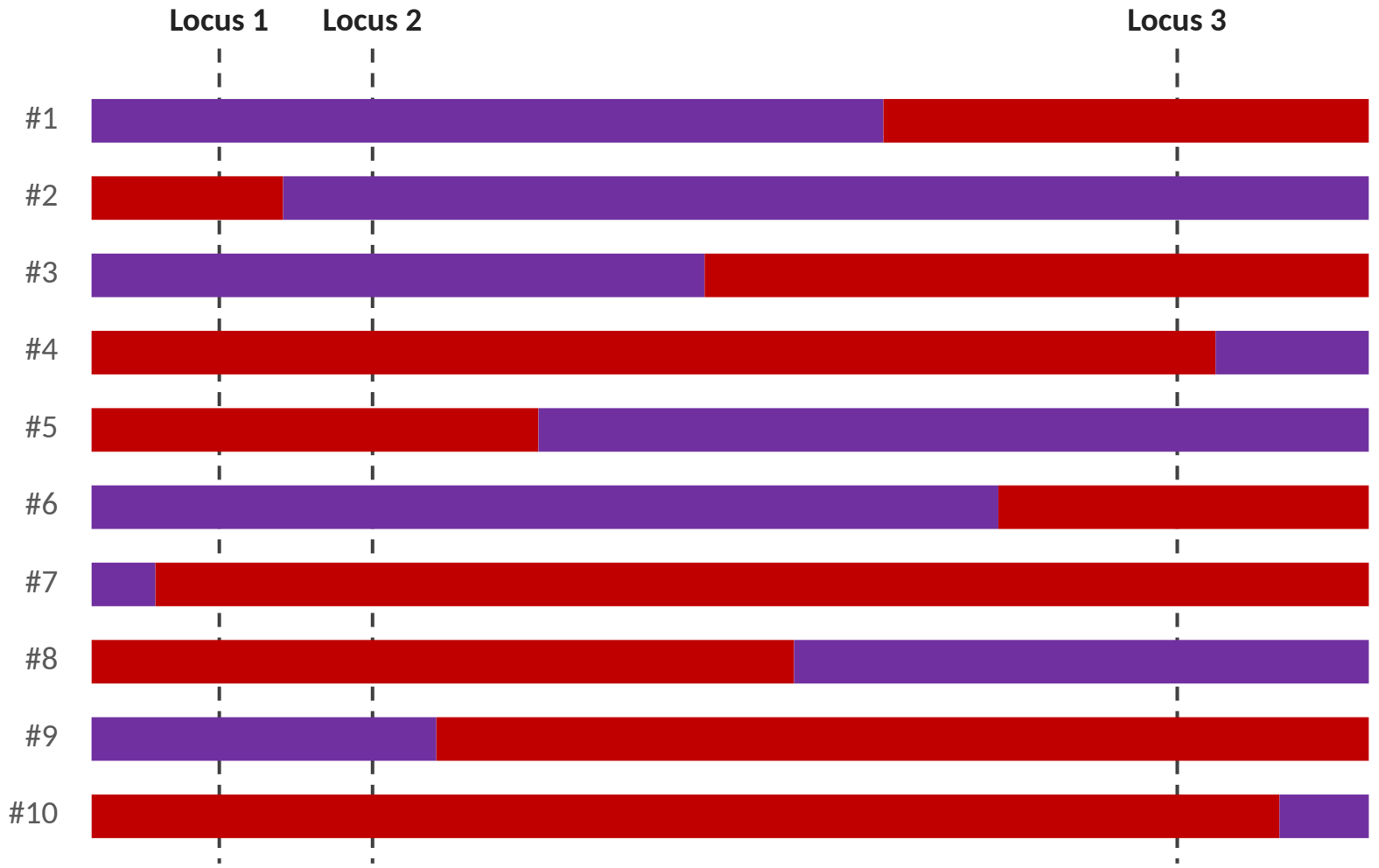


Haplotype: the set of alleles on one chromosome.
Recombination makes new haplotypes.



Loci far apart split up more often

10 meioses in the F1, one crossover each, at a random spot

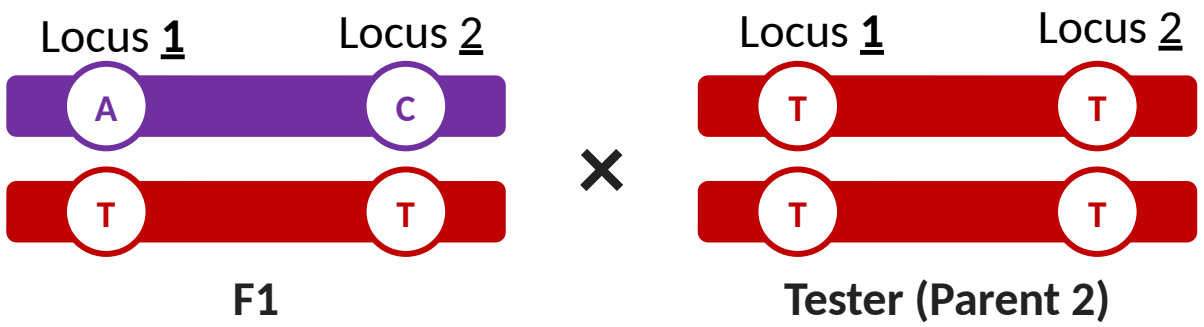


Loci 1 & 2 (close)
split apart in **1 of 10**

Loci 1 & 3 (far)
split apart in **7 of 10**

The trick: a testcross lets us read the F1's gametes

We can't watch crossovers, but we can genotype SNPs in the offspring



The tester can only give **T T**.
So each offspring's genotype tells us which gamete the F1 made.

Gamete from F1	+ from tester	Offspring SNPs Locus 1	Offspring SNPs Locus 2	Type
		**AT**	**CT**	Parental
		**TT**	**TT**	Parental
		**AT**	**TT**	**Recombinant**
		**TT**	**CT**	**Recombinant**

Your data: 20 testcross offspring

Each card = one offspring's SNP genotypes at Locus 1 and Locus 2. Which ones carry a recombinant chromosome from the F1?

#1 **AT CT**	#2 **TT TT**	#3 **AT CT**	#4 **AT TT**	#5 **TT TT**
#6 **TT TT**	#7 **AT CT**	#8 **AT CT**	#9 **TT CT**	#10 **TT TT**
#11 **AT CT**	#12 **TT TT**	#13 **TT CT**	#14 **AT CT**	#15 **TT TT**
#16 **AT CT**	#17 **TT TT**	#18 **AT TT**	#19 **TT TT**	#20 **AT CT**

Parental = **AT CT** or **TT TT** Recombinant = **AT TT** or **TT CT**

Live poll

Number: live histogram

How many of the 20 testcross offspring are recombinants?

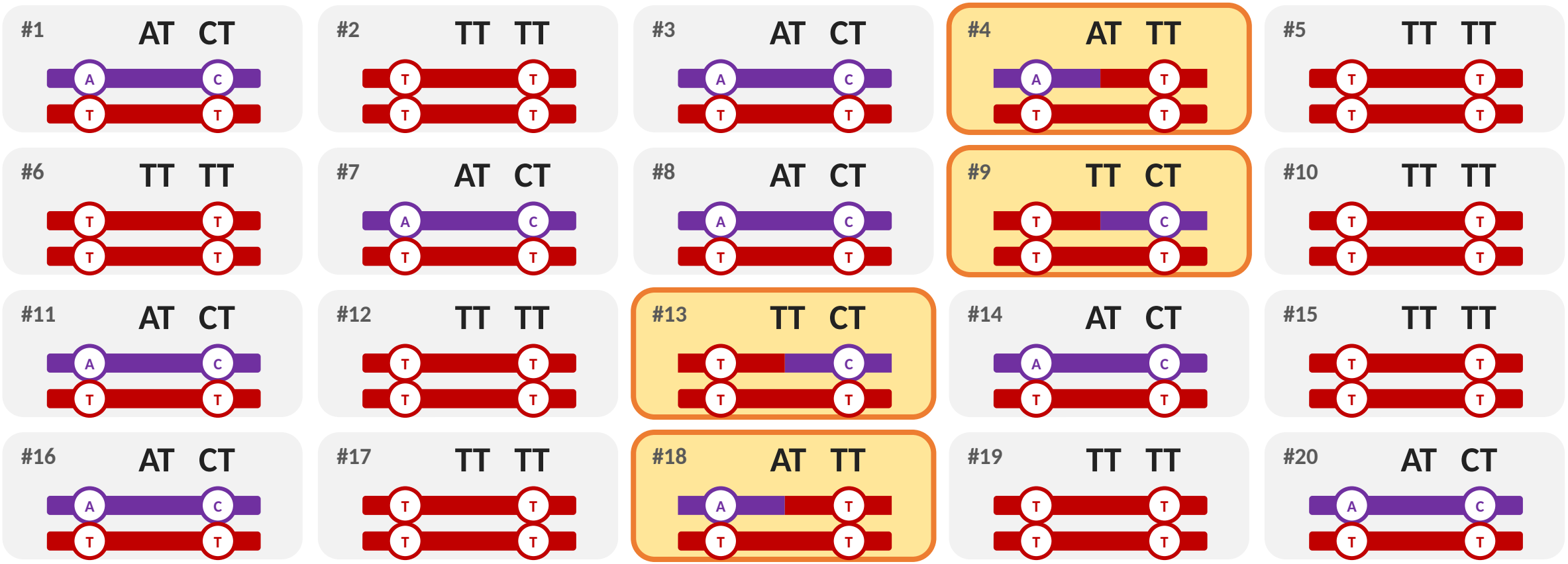
Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-rfcount

host screen

4 of 20 are recombinants

Each offspring is diploid: top chromosome from the F1, bottom (T T) from the tester



Recombination frequency = 4 / 20 = 20% → ~20 cM apart

Recombination frequency is a distance

Recombination frequency (RF) = recombinant offspring ÷ total offspring
1% recombination = 1 **centiMorgan (cM)** of genetic distance



~0%
Tightly linked: inherited together almost every time

20%
Our two SNPs: linked, ~20 cM apart

50%
Unlinked: looks exactly like Mendel's independent assortment

Four markers: what order are they in?

Recombination frequency (%) between each pair of markers

	A	B	C	D
A	—	20%	10%	15%
B	20%	—	30%	5%
C	10%	30%	—	25%
D	15%	5%	25%	—

Hint: the biggest number tells you which two markers are on the ends.

Live poll

Multiple choice

Using the table of recombination frequencies, what is the order of markers A, B, C, D along the chromosome?

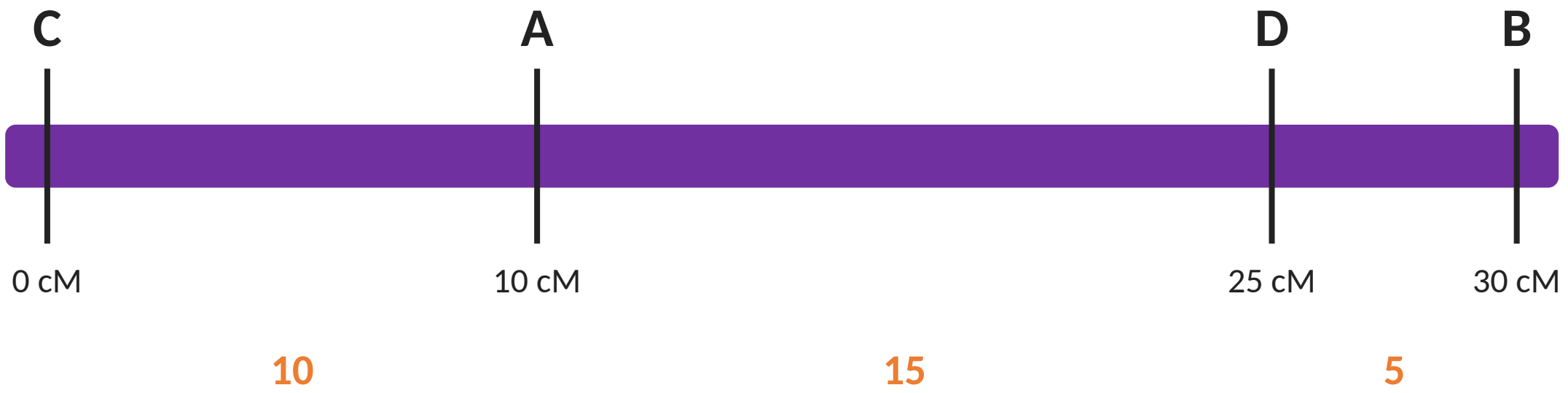
Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-maporder

host screen

A linkage map

Order C – A – D – B. Distances add up: 10 + 15 + 5 = 30



Repeat with hundreds of markers → a genetic map of every chromosome

Why the F2 is harder to read

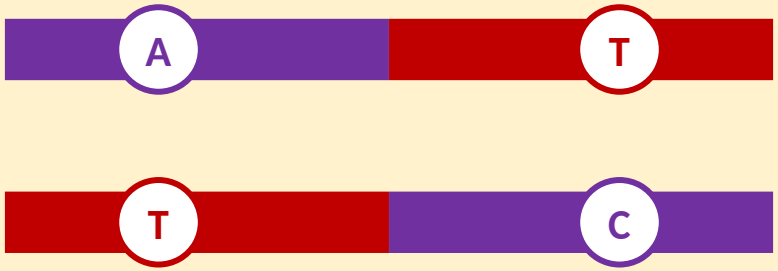
F2 offspring with SNPs AT CT : which story is true?

Story 1: two parental chromosomes



0 recombinants

Story 2: two recombinant chromosomes



2 recombinants

Same SNP genotype, different stories. That's why mapping uses crosses where one parent's chromosome is known (testcross, backcross) or inbred lines.

Why recombination matters

Finding genes

A marker that rarely recombines away from a trait sits near the gene.
This is how QTL mapping and GWAS work (coming later this quarter).

Breeding

A good allele drags its neighbors along ("linkage drag").
Recombination cuts off the bad hitchhikers.

Diversity

Every generation shuffles new haplotypes.
Why siblings differ, and why selection has new combinations to work with.

Play: the recombination game

Interactive linkage demo

Make crosses, collect offspring, count the recombinants, and build your own linkage map.

Scan with your phone

[Play on your phone or laptop: greymonroe.github.io/recombination-game/](greymonroe.github.io/recombination-game/)

Live poll

Text wall

Short Rht wheat always has many tillers. One gene doing two things (pleiotropy), or two tightly linked genes? How could you tell?

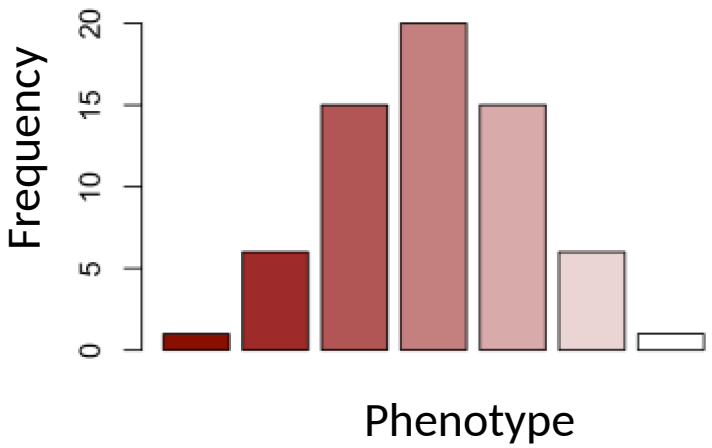
Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-pleiolink

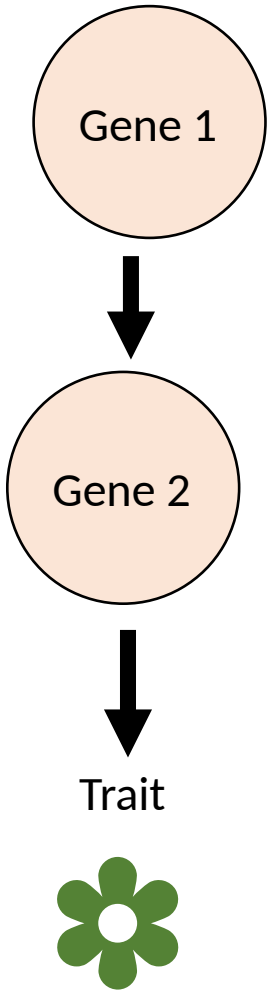
host screen

Most important take home messages

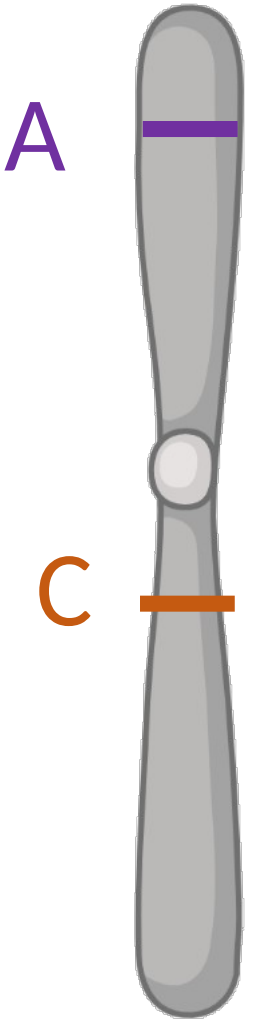
Incomplete dominance in many loci leads to **additive genetic variation** for **quantitative traits.**



Interactions between genes leads to **epistasis**



Physical linkage between loci on chromosomes leads to non-independent inheritance. **Close loci are Inherited together (recombine less).**



Next week



The Biology of Alleles